MTH104

Sets and Logic

Introduction

- What is this course?
- Why do we study it?
- What is the relation between sets and logic?

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Introduction

In mathematics, a proof is a demonstration that, assuming certain axioms, some statement is necessarily true. That is, proof is a logical argument, not an empirical one. One must demonstrate that a proposition is true in all cases before it is considered a theorem of mathematics. An unproven proposition for which there is some sort of empirical evidence is a conjecture. Mathematical logic is the framework upon which rigorous proofs are built. It is the study of the principles and criteria of valid inference and demonstrations.

Introduction

In set theory, we study sets, which are informally collection of objects. Although any type of object can be collected into a set, set theory is applied most often to objects that are relevant to mathematics. A rigorous analysis of set theory belongs to the foundations of mathematics and mathematical logic. A Set Theorist explores what is provable—and what is not provable—when one is allowed to use all of the usual tools of mathematical constructions and proofs (tools such as: creating

Introduction

function spaces, completions of various kinds of objects, taking quotients, forming products, induction and recursion, etc.).

Basic Concepts of Sets

Sets and Elements

A fundamental concept in all branches of mathematics is that of a set. A set is any well-defined list, collection, or class of objects. The objects in sets, as we shall see from our examples, can be anything: numbers, people, letters, rivers, etc. These objects are called the elements or members of the set.

Some Examples of Sets

1) The numbers 1, 3, 7 and 10.
2) The solution of the equation $x^2 - 3x - 2 = 0$.
3) The vowels of the english alphabet: a, e, i, o, u.
4) The people living on the earth.
5) The even integers 2, 4, 6, ...

Notation

Sets will usually be denoted by capital letters

A, B, X, Y,…

The elements in our sets will usually be represented by lower case letters

a, b, x, y,…

If we define a particular set by actually listing its members, for example, let A consists of the numbers 1,3, 7 and 10, then we write

$A = \{1, 3, 7, 10\}$ $1 \in A$ $2 \notin A$ belongs to

The statement "p belongs to A" is written as $p \in A$ .The statement that p is not an element of A, is written $p \notin A$

Notation

that is, the elements are separated by commas and enclosed in brackets { }. We call this the tabular form of a set.

But if we define a particular set by stating properties which its elements must satisfy, for example, let B be the set of all even numbers, then we use a letter, usually x, to represent an arbitrary element and we write

$B = \{x \mid x \text{ is even}\}$

We call this the set-builder form of a set.

Examples

1) Let $E = \{x \mid x^2 - 3x + 2 = 0\} = \{1, 2\}$
observe that $1 \in E$, $3 \notin E$.

2) $A = \{x \mid x \text{ is a vowel}\} = \{a, e, i, o, u\}$

3) Let $F = \{x \mid x \in P, 3 < x < 12\}$
where $P = \{1, 2, 3, 4, \ldots\}$
$F = \{4, 5, 6, 7, 8, 9, 10, 11\}$

Examples

Empty Set, Subsets and Universal Set

Universal Set

All sets under investigation in any application of set theory are assumed to be contained in some large fixed set called the universal set or universe. We will denote the universal set by U.

Examples are given below:

- In plane geometry, the universal set consists of all points in the plane.
- In human population studies the universal set consists of all people in the world.

Empty Set

It is convenient to introduce the concept of the empty set, that is, a set which contains no elements. This set is sometimes called the null set. We denote it by symbol ø.

Some Examples

1. Let A be the set of people in the world who are older than 200 years.
2. Let $B = \{x \mid x^2 = 4, x \text{ is odd}\}$. Then B is the empty set.
3. The set $S = \{x : x \text{ is a positive integer}, x^2 = 3\}$ has no elements since no positive integer has the required property.

$S = \phi$

Subsets

Suppose every element in a set A is also an element of a set B ; then A is called a subset of B. We also say that A is *contained in* B or B *contains* A. This relationship is written

$$A \subseteq B \quad or \quad B \supseteq A$$

$A = \{1, 2\}$, $B = \{1, 2, 3\}$

Examples

- Consider the sets

$A = \{1, 3, 5, 8, 9\}$, $B = \{1, 2, 3, 5, 7\}$, $C = \{1, 5\}$

Then $C \subseteq A$ and $C \subseteq B$. But $B \not\subset A$.

- The set $A = \{2, 3, 4, 5\}$ is not a subset of $B = \{x : x \text{ is even}\}$ since $3 \in A$ but $3 \notin B$.

$= \{2, 4, 6, \ldots\}$

Properties of Sets

- Every set A is a subset of the universal set U since, by definition, all the elements of A belong to U. Also empty set ø is a subset of A.
- Every set A is a subset of itself since, trivially, the elements of A belong to A. $A \subseteq A$
- If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
- If $A \subseteq B$ and $B \subseteq A$, then $A = B$.

Proper Subset

If $A \subseteq B$, then it is still possible that $A = B$. When $A \subseteq B$ but $A \neq B$, we say that A is a proper subset of B.

For example, suppose

$A = \{1,2\}$, $B = \{1,2,3\}$, $C = \{1,3,2\}$

Then A and B are both subsets of C; but A is a proper subset of C, whereas B is not a proper subset of C.

Set Operations

In this topic, we will study a number of set operations, including the basic operations of union, intersection, and difference of sets, where new sets will be assigned to pairs of sets A and B.

Union and Inersection

The union of two sets A and B, denoted by $A \cup B$, is the set of all elements which belong to A or B; that is

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

The intersection of two sets A and B, denoted by $A \cap B$, is the set of all elements which belong to both A and B; that is

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

Example 1

Let $A = \{1,2,3,4\}$, $B = \{3,4,5,6,7\}$, $C = \{2,3,8,9\}$. Then

$A \cup B = \{1,2,3,4,5,6,7\}$, $A \cap B = \{3,4\}$

$A \cup C = \{1,2,3,4,8,9\}$, $A \cap C = \{2,3\}$

$B \cup C = \{3,4,5,6,7,2,8,9\}$, $B \cap C = \{3\}$

$= \{2,3,4,5,6,7,8,9\}$

Example 2

Let U denote the set of students at a university, and let M and F denote, respectively, the set of male and female students at the university. Then

$$M \cup F = U$$

since each student in U is either in M or in F. On the other hand,

$$M \cap F = \varnothing$$

since no students belong to both M and F.

Some Properties of Union and Intersection

For any set A and B, we have

1) $A \cap B \subseteq A$ and $A \cap B \subseteq B$
2) $A \subseteq A \cup B$ and $B \subseteq A \cup B$

By combining the above two results, we can write

$$A \cap B \subseteq A \subseteq A \cup B \text{ and } A \cap B \subseteq B \subseteq A \cup B$$

Complement

Complement of a set A, denoted by A^c, is the set of elements which belong to U but which do not belong to A; that is,

$$A^c = \{x : x \in U, x \notin A\}$$

Example 3

Let $U = \{a, b, c, \ldots, y, z\}$ be the universal set, and let $A = \{a,b,c,d,e\}$, $B = \{e, f, g\}$, $V = \{a,e,i,o,u\}$

Then

$A^c = \{f, g, h, \ldots, y, z\}$

$B^c = \{a, b, c, d, h, i, \ldots, y, z\}$

V^c = consonants

Example 4

Suppose the set R of real numbers is the universal set. $\mathbb{Q}$ denotes the set of rational numbers. Hence $\mathbb{Q}^c$ will denote the set of irrational numbers.

$\mathbb{Q} = \{\frac{p}{q}, p \in \mathbb{Z}, q \in \mathbb{Z}, q \neq 0\}$ $\{\sqrt{2}, \sqrt{3}\}$ irrational

$\mathbb{Q}^c = ?$

Difference of Sets

Let A and B be sets. The difference of A and B, denoted by A/B, is the set of elements which belong to A but which do not belong to B ; that is,

$$A / B = \{x : x \in A, x \notin B\}$$

Many texts denote A/B by A-B.

Example 5

Consider the sets

$A = \{1,2,3,4\}$, $B = \{3,4,5,6,7\}$, $C = \{6,7,8,9\}$

Then

$A/B = \{1,2\}$ ✓

$B/C = \{3,4,5\}$

$B/A = \{5,6,7\}$ ✓

$C/B = \{8,9\}$

Symmetric Difference

The symmetric difference of the sets A and B, denoted by

$A \oplus B = (A \cup B) / (A \cap B)$

or

$A \oplus B = (A/B) \cup (B/A)$

Example 6

Consider the sets

$A = \{1,2,3,4\}$, $B = \{3,4,5,6,7\}$, $C = \{6,7,8,9\}$

Then

$A \oplus B = (A \cup B) / (A \cap B) = \{1,2,3,4,5,6,7\} / \{3,4\}$

$= \{1,2,5,6,7\}$

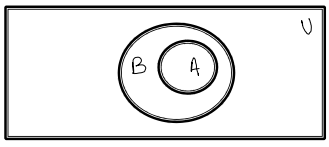
$B \oplus C = \{3,4,5,6,7,8,9\} / \{6,7\}$

$= \{3,4,5,8,9\}$

Venn Diagrams

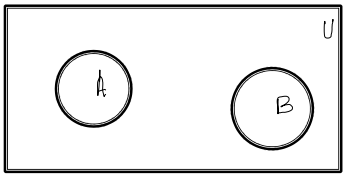
Venn Diagrams

A Venn diagram is a pictorial representation of sets where sets are represented by enclosed areas in the plane. The universal U represented by the points in a rectangle, and other sets are represented by disks lying within the rectangle. If $A \subseteq B$, then the disks representing A will entirely within the disk representing B, as shown in the following figure.



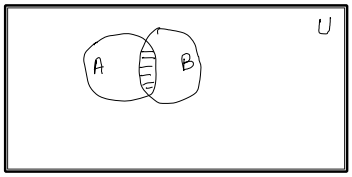
Venn Diagrams

If A and B are disjoint, then the disk representing A will be separated from the disk representing B, as in the following figure.



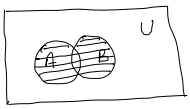
Venn Diagrams

On the other hand, if A and B are two arbitrary sets, it is possible that some elements are in A but not in B, some elements are in B but not in A, some are in both A and B, and some are in neither A nor B; hence , in general, we represent A and B as in the following way.

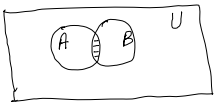


Some Examples

1) Union of A and B

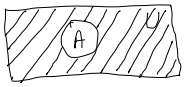


2) Intersection of A and B



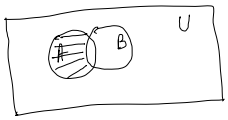
Examples

3) Complement of A



$A^c = U - A$

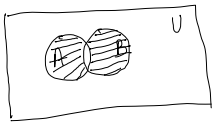
4) Difference of A and B



$A - B = \{x \mid x \in A, x \notin B\}$

Examples

7) Symmetric difference of A and B



$A \oplus B = (A \cup B)/(A \cap B)$

Algebra of Sets, Duality

Laws of the Algebra of Sets

Sets under the operations of union, intersection and complement satisfy various laws(identities). Which are listed below one by one.

- Idempotent Laws

$$A \cup A = A \text{ and } A \cap A = A$$

$A \cup A = A$

L.H.S $= A \cup A$
$= \{x : x \in A \text{ or } x \in A\}$
$= \{x : x \in A\} = A$

$A \cap A = A$

L.H.S $= A \cap A$
$= \{x : x \in A \text{ and } x \in A\}$
$= \{x : x \in A\} = A$

Laws of the Algebra of Sets

- Associative Laws

$$A\cup(B\cup C)=(A\cup B)\cup C$$

and

$$A\cap(B\cap C)=(A\cap B)\cap C$$

$A\cup(B\cup C)=(A\cup B)\cup C$

L.H.S $=A\cup(B\cup C)$

$=\{x: x\in A \text{ or } x\in(B\cup C)\}$

$=\{x: x\in A \text{ or } x\in B \text{ or } x\in C\}$

$=\{x: x\in A\cup B \text{ or } x\in C\} = (A\cup B)\cup C$

Laws of the Algebra of Sets

- Commutative Laws

$$A\cup B=B\cup A$$

and

$$A\cap B=B\cap A$$

$A\cup B=\{x: x\in A \text{ or } x\in B\}$

$=\{x: x\in B \text{ or } x\in A\}$

$=B\cup A$

$A\cap B=\{x: x\in A \text{ and } x\in B\}$

$=\{x: x\in B \text{ and } x\in A\}$

$=B\cap A$

Laws of the Algebra of Sets

- Distributive Laws

$$A\cup(B\cap C)=(A\cup B)\cap(A\cup C)$$

and

$$A\cap(B\cup C)=(A\cap B)\cup(A\cap C)$$

L.H.S $=A\cap(B\cup C)$

$=\{x: x\in A \text{ and } x\in B\cup C\}$

$=\{x: x\in A \text{ and } x\in B \text{ or } x\in A \text{ and } x\in C\}$

$=\{x: x\in(A\cap B) \text{ or } x\in(A\cap C)\}$

$=(A\cap B)\cup(A\cap C)$

Laws of the Algebra of Sets

- Identity Laws

$$A\cup\phi=A,\ A\cap U=A$$

and

$$A\cup U=U,\ A\cap\phi=\phi$$

L.H.S $=A\cup\phi=\{x: x\in A\}=A=$ R.H.S

R.H.S $=A=\{x: x\in A\}=\{x: x\in A \text{ and } x\in U\}$

$=A\cap U$

Laws of the Algebra of Sets

- Involution Law

$$(A^c)^c=A$$

Note that $A^c=\{x: x\in U, x\notin A\}$

so, $(A^c)^c=\{x: x\in U, x\notin A^c\}$

$=\{x: x\in U, x\in A\}$

$=A$

Laws of the Algebra of Sets

- Complement Laws

$$A\cup A^c=U,\ A\cap A^c=\phi$$

and

$$U^c=\phi,\ \phi^c=U$$

1) L.H.S $=A\cup A^c$

$=\{x: x\in A \text{ or } x\in A^c\}$

$=\{x: x\in A \text{ or } x\in U, x\notin A\}$

$=\{x: x\in U\}=U=$ R.H.S

2) L.H.S $=A\cap A^c$

$=\{x: x\in A \text{ and } x\in A^c\}$

$=\{x: x\in A \text{ and } x\in U, x\notin A\}$

$=\phi=$ R.H.S

Laws of the Algebra of Sets

- DeMorgan's Laws

1) $(A \cup B)^c = A^c \cap B^c$

and

2) $(A \cap B)^c = A^c \cup B^c$

1) L.H.S $= (A \cup B)^c$
$= \{x : x \notin (A \text{ or } B)\}$
$= \{x : x \notin A \text{ and } x \notin B\}$
$= \{x : x \in A^c \text{ and } x \in B^c\}$
$= A^c \cap B^c = R.H.S$

Duality

The identities that we have discussed above are arranged in pairs. We now consider the principle behind this arrangement. Let E be an equation of set algebra. The dual E^* of E is the equation obtained by replacing each occurrence of $\cup, \cap, U, \varnothing$ in E by $\cap, \cup, \varnothing, U$, respectively.

For example, the dual of

$(U \cap A) \cup (B \cap A) = A$ is $(\varnothing \cup A) \cap (B \cup A) = A$

dual of

Example

Question. Write the dual $(A \cap U)^c \cap A = \varnothing$.

Solution.

The dual is

$(A \cup \phi)^c \cup A = U$

Finite Sets, Counting Principles

Finite Sets

A set is said to be finite if it contains exactly m distinct elements where m denotes some nonnegative integer. Otherwise set is said to be infinite. For example, the empty set ø and the set of letters of the English alphabet are finite sets, whereas the set of even positive integers $\{2, 4, 6, \ldots\}$ is infinite.

Notation

The notation $n(A)$ or $|A|$ will denote the number of elements in a finite set A.

Lemma 1

Suppose A and B are finite disjoint sets. Then $A \cup B$ is finite and

$$n(A \cup B) = n(A) + n(B)$$

For example,

$A = \{1, 2, 3\}$ and $B = \{a, b, c\}$

Then,

$A \cup B = \{1, 2, 3, a, b, c\}$

$n(A \cup B) = 6 = n(A) + n(B) = 3 + 3 = 6$

Theorem 1

Let A be any set in a finite universal set U. Then

$$n(A^c) = n(U) - n(A)$$

For example, if there are 20 male students in a class of 35 students, then there are 35-20=15 female students.

$n(U) - n(A)$

Theorem 2

Suppose A and B are finite sets. Then

$$n(A \setminus B) = n(A) - n(A \cap B)$$

For example, suppose an archery class A contains 35 students, and 15 of them are also in a bowling class B. Then

$$n(A \setminus B) = n(A) - n(A \cap B) = 35 - 15 = 20$$

That is, there are 20 students in the class A who are not in class B.

Theorem 3

Suppose A and B are finite sets. Then

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

For example,

$A = \{2, 4, 6, 8\}$, $B = \{6, 8, 10, 12\}$

Then, $A \cup B = \{2, 4, 6, 8, 10, 12\}$

$n(A \cup B) = 6$

$= n(A) + n(B) - n(A \cap B)$

$= 4 + 4 - 2 = 6$

$A \cap B = \{6, 8\}$

Corollary 1

Suppose A, B, C are finite sets. Then $A \cup B \cup C$ is finite and

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C).$$

Example

Consider the following data among 110 students in a college dormitory:

30 students are on a list A (taking Accounting),
35 students are on a list B (taking Biology),
20 students are on both lists.

Find the number of students:

(a) On list A or B

(b) On exactly one of the two lists,

(c) On neither list.

Example

a) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 30 + 35 - 20 = 45$ ✓

b) $n(A / B) = n(A) - n(A \cap B)$

$= 30 - 20 = 10$

$n(B / A) = n(B) - n(A \cap B)$

$= 35 - 20 = 15$

c) $n(A^c \cap B^c) = n((A \cup B)^c)$

$= n(U) - n(A \cup B)$

$= 110 - 45$

$= 65$

$(A \cup B)^c = A^c \cap B^c$

$\therefore n(A^c) = n(U) - n(A)$

Classes of Sets, Power Sets

Classes of Sets

Given a set S, we may wish to talk about some of its subsets. Thus we would be considering a "set of sets". Whenever such a situation arises, to avoid confusion, we will speak of *class* of sets or a *collection* of sets. If we wish to consider some of the sets in a given class of sets, then we will use the term *subclass* or *subcollection*.

Classes of Sets

To understand this concept, we consider the set set $S = \{1, 2, 3, 4\}$. Let A be the class of subsets of S which contain exactly three elements of S. Then

$A = [\{1,2,3\}, \{1,2,4\}, \{1,3,4\}, \{2,3,4\}]$

Let B be the class of subsets of S which contain 2 and two other elements of S. Then

$B = [\{2,1,3\}, \{2,1,4\}, \{2,3,4\}]$

Here we can see that B is a subclass of A.

Power Sets

For a given set S, we may speak about the class of all subsets of S. This class is called the power set of S, and it will be denoted by $P(S)$. If S is finite, then so is $P(S)$.

Also

$$n(P(S)) = 2^{n(S)}$$

Power Sets

Suppose $S = \{1,2,3\}$. Then

$P(S) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$

Note that

$2^{n(S)} = 2^3 = 8$ elements

Mathematical Induction

Introduction

Suppose we have an infinite ladder, as shown in Figure 1.

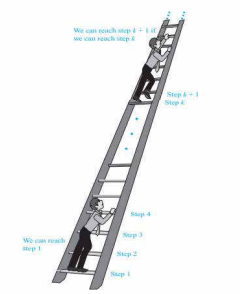


FIGURE 1 Climbing an Infinite Ladder.

Introduction

We want to know whether we can reach every step on this ladder. We know two things:

1. We can reach the first step of the ladder.
2. If we can reach a particular step of the ladder, then we can reach the next step.

Can we conclude that we can reach every step?

Answer is yes. Something we can verify using an important proof technique called mathematical induction.

Mathematical Induction

In general, mathematical induction can be used to prove statements that assert that $P(n)$ is true for all positive integers n, where $P(n)$ is a proposition. A proof by mathematical induction has two parts, a basis step, where we show that $P(1)$ is true, and an inductive step, where we show that for all positive integers k, if $P(k)$ is true, then $P(k+1)$ is true.

Principle of Mathematical Induction

To prove that $P(n)$ is true for all positive integers n, where $P(n)$ is a proposition, we complete two steps:

Basis Step: We verify that $P(1)$ is true.

Inductive Step: We show that the conditional statement $P(k) \longrightarrow P(k+1)$ is true for all positive integers k.

If $P(k)$ is true then we will prove that $P(k+1)$ is true.

Example 1

Show that if n is a positive integer, then

$$1 + 2 + \ldots + n = \frac{n(n+1)}{2}$$

$1+2+3 = \frac{3(3+1)}{2} = \frac{12}{2} = 6$

Let $P(n)$ be the proposition. First we show that $P(1)$ is true and then conditional statement $P(k)$ implies $P(k+1)$ is true for $k = 1, 2, 3, \ldots$.

~~Basis step~~

$P(1)$ is true: $1 = \frac{1(1+1)}{2}$

$1 = \frac{2}{2} = 1$

Example 1

Inductive step: Assume that $P(k)$ is true for an arbitrary positive integer k

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}$$ ✓

Under this assumption it must be shown that $P(k+1)$ is true, namely

$$1 + 2 + \cdots + k + (k+1) = \frac{(k+1)[(k+1)+1]}{2} = \frac{(k+1)(k+2)}{2}$$

Now, $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$

$$1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1)$$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

Example 1

$$= \frac{k(k+1) + 2(k+1)}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

This last equation shows that $P(k+1)$ is true under the assumption that $P(k)$ is true

Example 2

Let $P(n)$ be the assertion that the sum of the first n odd integers is n^2; that is,

$P(n)$: $1+3+5+\ldots+(2n-1)=n^2$.

$1+3+5 = 3^2$

$9 = 9$

Basis Step: $P(1)$ is true, because

$1 = 1^2$

Inductive Step: Assume that $P(k)$ is true. That is,

$1+3+5+\cdots+(2k-1) = k^2$

Next odd number after $2k-1$ is $2k+1$.

Example 2

$1+3+\cdots+(2k-1) = k^2$

$1+3+\cdots+(2k-1)+(2k+1) = k^2+(2k+1)$

$= k^2+2k+1$

$= (k+1)^2$

That is, $P(k+1)$ is true whenever $P(k)$ is true.

Proved

Example 2

Example 3

Use mathematical induction to show that

$P(n)$: $1+2^1+2^2+\ldots+2^n = 2^{n+1}-1$

for all nonnegative integers n.

$2^{(k+1)+1}-1$

$= 2^{k+2}-1$

Basis Step: Let $P(n)$ be the statement

$P(0)$ is true, because

$2^0 = 2^{0+1}-1$

$1 = 2^1-1$

$1 = 1$

Inductive step: Assume that $P(k)$ is true. That $1+2+2^2+\cdots+2^k = 2^{k+1}-1$

Example 3

Axiomatic Development of Set Theory

What is axiom and axiomatic development?

Any axiomatic development of a branch of mathematics begins with the following:

1) undefined terms,
2) Undefined relations,
3) Axioms relating the undefined terms and undefined relations.

Then one develops theorems based upon the axioms and definitions.

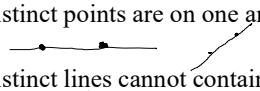
Axiomatic Development of Eulidean Geometry

Consider, for example, the axiomatic development of plane Euclidean geometry. It begins with the following:

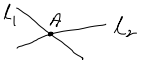
1) "points" and "lines" are undefined terms;
2) "point on a line" is an undefined relation.

Two of many axioms of Euclidean geometry follow:

Axiom 1: Two distinct points are on one and only one line.



Axiom 2: Two distinct lines cannot contain more than one point in common.



Axiomatic Development of Set Theory

The axiomatic development of set theory begins with the following:

1) "element" and "set" are undefined terms;
2) "elements belong to a set" is the undefined relation.

Two of the axioms of set theory follow:

Principle of Extension: Two sets A and B are equal if and only if they have the same elements.

Axiomatic Development of Set Theory

Principle of Abstraction: Given any set U and any property P, there is a set A such that the elements of A are exactly those elements in U which have the property P.

$U = \{$Nauman, Sara, Aleesha, Mansoor$\}$

$A = \{$Nauman, Mansoor$\}$

Real Number System

Real Number System

The notation $\underline{R}$ will be used to denote the real numbers. These are the numbers one uses in basic arithmetic and algebra. R together with its properties is called the real number system.

The set R of real numbers includes the following sets of numbers:

$Z = \{\ldots, -2, -1, 0, 1, 2, 3, \ldots\}$ = set of integers

$Z^+ = P = \{1, 2, 3, \ldots\}$ = set of positive integers

$N = \{0, 1, 2, 3, \ldots\}$ = set of nonnegative integers

Q = set of rational numbers i.e. numbers which are ratios of integers $\frac{2}{3}, \frac{-1}{2}, \frac{2}{1} = 2$

Real Number System

Irrational Numbers

Those real numbers which cannot be represented as the ratio of integers, are called irrational numbers.

For example, $\pi = 3.14\ldots \neq \frac{22}{7}$

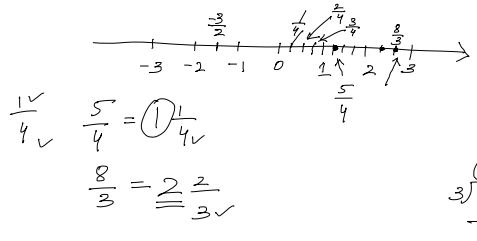
$\sqrt{2}$

$\sqrt{3}$

$R = Q \cup Q'$

Real Line *R*, Decimal Expansion

One of the most important properties of the real numbers is that they can be represented graphically by points on a straight line.



$\frac{1}{4}$ ✓ $\frac{5}{4} = 1\frac{1}{4}$ ✓

$\frac{8}{3} = 2\frac{2}{3}$ ✓

$4\overline{)5}$ ① 4 ②

$3\overline{)8}$ 6 2

Decimal Expansion

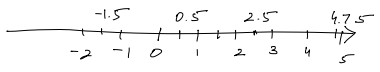
Real numbers can also be represented by decimals. The decimal expansion of a rational number will either stop as in $\frac{3}{4} = 0.75$ or will have a pattern that repeats indefinitely, such as $1\frac{7}{11} = 1.\overline{54}$ $\overline{54}\ldots$ Even when the decimal expansion stops, it can be rewritten using repeated 9`s, for example, $\frac{3}{4} = 0.74999\ldots$.

$= 0.75$, $0.3 = 0.2999\ldots$

The decimal expansion of an irrational number never stops nor does it have a repeating pattern. The points representing the decimal 2.5 and 4.75 are indicated below on real line.

$\pi = 3.14\ldots$

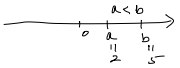
Decimal Expansion



Order and Inequalities

Order Relation of Real Numbers

Let a and b be real numbers. We say a is less than b, written $a < b$ if the difference $b-a$ is positive. $2 < 5$, $5-2$ is positive. Geometrically, $a < b$ if and only if the point a lies to the left of the point b on the real line R.



Observe that we define order in R in terms of the positive real numbers denoted by R^+.

Main Properties of Positive Real Number

There are two main properties of the positive real numbers R^+:

1) If a and b are positive, then $a+b$ and ab are positive.
2) For any real number a, either a is positive, $a = 0$, or $-a$ is positive. $a = 2$, $a = -2$; $-a = -(-2) = 2$, $a = 0$

The following additional notation and terminology are used:

$a > b$, means b < a

$a \leq b$, means $a < b$ or $a = b$ $2 \leq 2$ or $2=2$

$a \geq b$, means $b \leq a$ $2 < 5$ or $5 > 2$.

Inequality

Any statement of the form $a < b$, $a \leq b$, $a > b$, or $a \geq b$ is called inequality; and any statement of the form $a < b$ or $a > b$ is sometimes called a strict inequality.

$2 < 7$ ✓ strict inequality

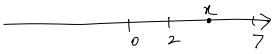
$a \geq b$ ✗ not strict inequality

Example 1

a) $2 < 5$; $-6 < -3$, $4 \leq 4$, $5 > -8$, $b \geq 0$, $-7 \leq 0$

b) A real number a is positive iff $a > 0$, and a is negative iff $a < 0$.

c) The statement $2 < x < 7$ means $2 < x$ and $x < 7$.



Properties of Inequality

Basic properties of the inequality relation follow.

1) Let a, b, c be real numbers. Then:
   (i) $a \leq a$ $2 \leq 2$
   (ii) If $a \leq b$ and $b \leq a$, then $a = b$. $3 \leq 3$, $\Rightarrow 3 = 3$
   (iii) If $a \leq b$ and $b \leq c$, then $a \leq c$. $2 < 5$, $5 < 7$ $\Rightarrow 2 < 7$
2) For any real numbers a and b, exactly one of the following holds:
   $a < b$, $a = b$ or $a > b$

Properties of Inequality

3) Let a, b, c be real numbers such that $a \leq b$. Then
   (i) $a+c \leq b+c$.
   (ii) $ac \leq bc$ when $c > 0$; but $ac = bc$ when $c = 0$.

$a = 2$, $b = 5$, $c = 2$

$a < 5 = b$

$a + c < b + c$

$4 < 7$

$a < b$

$ac < bc$

$4 < 10$

Let $c = -2$

$ac > bc$

$-4 > -10$

Properties of Inequality

Q. Prove that if $a \le b$ and $b \le c$, then $a \le c$.

Proof: If $a = b$ and $b = c$, then $a = c$.

If $a < b$ and $b < c$, then $b - a$ is positive and $c - b$ is positive.

Now, $(b - a) + (c - b)$ is positive.

$= -a + c$

$= c - a$ is positive

$\Rightarrow a < c$

Hence, $a \le c$. Proved

Properties of Inequality

Q. Let a, b, c be real numbers. Then prove that (and suppose $a \le b$)

(i) $a + c \le b + c$.

(ii) $ac \le bc$ when $c > 0$; but $ac \ge bc$ when $c < 0$.

Do it yourself as assignment.

If $a = b$, then $ac = bc$

If $a < b$, then $b - a$ is positive

$c(b - a)$ is also positive.

$= bc - ac$ is " "

$\Rightarrow ac < bc$

Hence $ac \le bc$

Absolute Value, Distance

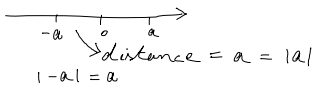
Absolute value of a Number

The absolute value of a real number a, denoted by $|a|$, may be viewed as the distance between a and the 0 on the real line R. Formally, $|a| = a$ or $-a$ according as a is positive or negative, and $|0| = 0$. That is:

$$|a| = \begin{cases} a, & \text{if } a \ge 0 \\ -a, & \text{if } a < 0 \end{cases}$$

Accordingly, $|a|$ is always positive when $a \neq 0$.

Geometrically, $|a|$ is:



$|-a| = a$

Example 1

$|-3| = -(-3) = 3$, $|2| = 2$, $|-3.5| = -(-3.5) = 3.5$

$|4.5| = 4.5$

The Distance Between Two Numbers

The distance d between two points a and b is denoted by $d(a, b)$ and is obtained from the formula

$$d = d(a, b) = |a - b| = |b - a|$$

For example,

2, 5

$d(2, 5) = |2 - 5| = |-3| = 3$

-3, 11

$d(-3, 11) = |-3 - 11| = |-14| = 14$

Properties of Absolute Value Function

Let a and b be any real numbers.

1) $|a| \geq 0$, and $|a| = 0$ iff $a = 0$.
2) $-|a| \leq a \leq |a|$.
3) $|ab| = |a||b|$. ✓
4) $|a \pm b| \leq |a| + |b|$. ✓
5) $||a| - |b|| \leq |a \pm b|$. ✓

$a = -5$, $|a| = 5$, $\Rightarrow a \leq |a|$
$-|a| = -5 \leq a$
$\Rightarrow -|a| \leq a$

$a = 2$, $b = -3$
$ab = -6$
$|ab| = 6$
$|a| = 2, |b| = 3$
$|a||b| = 6$

Let $a = 5$, $b = -2$
$|a + b| = |5 - 2| = 3$ ✓
$|a| = 5$, $|b| = 2$
$|a| + |b| = 7$
$3 < 7$
$|a + b| < |a| + |b|$

Properties of Absolute Value Function

Q. Prove that $|ab| = |a||b|$. where a and b are real nos. ✓

Proof: 1) Suppose $a = 0$ and $b = 0$, then $|ab| = 0 = |a||b|$

2) Suppose $a > 0$, $b > 0$, then $ab > 0$
$\Rightarrow |a| = a$, $|b| = b$, $|ab| = ab$
$\Rightarrow |ab| = ab = |a||b|$

3) Suppose $a > 0$ and $b < 0$, then $ab < 0$.
$\Rightarrow |a| = a$, $|b| = -b$, $|ab| = -ab$
$\Rightarrow |ab| = -ab = a(-b) = |a||b|$

4) Suppose $a < 0$, $b > 0$ (Do it yourself)

5) Suppose $a < 0$ and $b < 0$, then $ab > 0$.
$\Rightarrow |a| = -a$, $|b| = -b$, $|ab| = ab$
$\Rightarrow |ab| = ab = (-a)(-b) = |a||b|$

Properties of Absolute Value Function

Q. Prove that $|a \pm b| \leq |a| + |b|$. ✓

Proof: 1) $|a + b| \leq |a| + |b|$ 2) $|a - b| \leq |a| + |b|$

$(a+b)^2 = a^2 + 2ab + b^2 \leq a^2 + 2|a||b| + b^2$
$= |a|^2 + 2|a||b| + |b|^2$
$= (|a| + |b|)^2$
$\Rightarrow (a+b)^2 \leq (|a| + |b|)^2$
$\Rightarrow \sqrt{(a+b)^2} \leq \sqrt{(|a| + |b|)^2}$ $\because \sqrt{x^2} = |x|$
$\Rightarrow |a + b| \leq |a| + |b|$ Proved

Note that
$ab \leq |ab| = |a||b|$
$\Rightarrow ab \leq |a||b|$
$\Rightarrow 2ab \leq 2|a||b|$

Also, $|a - b| = |a + (-b)|$
$\leq |a| + |-b|$
$= |a| + |b|$

Intervals

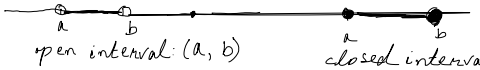
Intervals

Let a and b be distinct real numbers with, say, $a < b$. The intervals with endpoints a to b are denoted and defined as follows:

$(a, b) = \{x: a < x < b\}$, open interval from a to b
$[a, b] = \{x: a \leq x \leq b\}$, *closed* interval from a to b
$[a, b) = \{x: a \leq x < b\}$, *closed*-open interval from a to b
$(a, b] = \{x: a < x \leq b\}$, open-closed interval from a to b

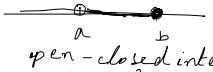
Intervals

- How do we represent these intervals on a real line?

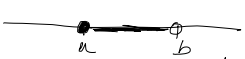


open interval: (a, b)

closed interval: $[a, b]$



open-closed interval
$(a, b]$



closed-open interval
$[a, b)$

Example 1

Find the interval satisfying each inequality, i.e., rewrite the inequality in terms of x alone:

1) $2 \le x-5 \le 8$

Add 5 throughout the inequality:

$2+5 \le x-5+5 \le 8+5$

$7 \le x \le 13$

$[7, 13]$

2) $6 \le -2x \le 4$

Multiply $-\frac{1}{2}$ throughout the inequality

$-\frac{1}{2} \times 6 \ge -\frac{1}{2} \times -2x \ge -\frac{1}{2} \times 4$

$-3 \ge x \ge -2$,

or $-2 \le x \le -3$ or $[-3, -2]$

Example 2

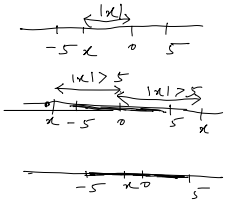
The inequality $|x| < 5$ may be interpreted to mean that the distance between x and the origin 0 is less than 5; hence x must lie between -5 and 5 on the real line R. How?

$|x| < a$

$\Rightarrow -a < x < a$

$|x| < 10$

$\Rightarrow -10 < x < 10$



Example 3

Q. Rewrite $|x-5| < 5$ without the absolute value sign.

Sol:

$|x-5| < 5$

$\Rightarrow -5 < x-5 < 5$

$\Rightarrow -5+5 < x-5+5 < 5+5$

$\Rightarrow 0 < x < 10$ Ans.

$(0, 10)$

Definition

A set A of real numbers is said to be dense in R if every open interval contains a point of A.

Example 1: The rational numbers Q are dense in R.

Like $(2, 4)$, Let's say $3 \in (2, 4)$ and $3 \in Q$

$(\frac{1}{3}, \frac{1}{2})$, Here $\frac{1}{4} \in (\frac{1}{3}, \frac{1}{2})$ and $\frac{1}{4} \in Q$

Example 2: The integers Z is not dense in R.

For example, Take $(\frac{1}{3}, \frac{1}{2})$

We cannot find, point on $(\frac{1}{3}, \frac{1}{2})$

such that it is a member of Z.

Infinite Intervals

Let a be any real number. Then the set of real numbers x satisfying $x < a$, $x \le a$, $x > a$, *or* $x \ge a$, is called an infinite interval with endpoint a. The interval is said to be closed or open according as the endpoint a does or does not belong to the interval. The four infinite intervals may also be denoted and defined as follows:

open interval: $x < a : (-\infty, a)$; $x \le a : (-\infty, a]$

$x > a : (a, +\infty)$; $x \ge a : [a, +\infty)$

↖ closed intervals

Infinite Intervals

These infinite intervals are pictured as:

$(-\infty, a)$:

(a, ∞) :

$(-\infty, a]$

$[a, \infty)$

Bounded Sets,Completion Property

Bounded Set

Let A be a set of real numbers. Then A is said to be bounded according as there exists a real number M such that, for every $x \in A$, we have $|x| \leq M$. $-M \leq x \leq M$

The number M is called a bound. Note that A is bounded if and only if A is a subset of some finite interval. Specifically M is a bound of A if and only if A is a subset of $[-M, M]$. ✓

$|x| \leq a \Rightarrow -a \leq x \leq M$

For example,

$A = \{1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{n}, \ldots\}$ is bounded.

Set A is a subset of the closed unit interval $I = [0,1]$. i.e $A \subseteq [0,1]$.

Bounded From Above

Let A be a set of real numbers. Then A is said to be bounded from above according as there exists a real number M such that, for every $x \in A$, we have $x \leq M$.

The number M is called an upper bound.

For example, $C = \{\ldots, -5, -3, -1\}$ is bounded from above. Here upper bound is $M = -1$. since for all $x \in C$, we have $x \leq M = -1$.

Bounded From Below

Let A be a set of real numbers. Then A is said to be bounded from below according as there exists a real number M such that, for every $x \in A$, we have $x \geq M$.

The number M is called a lower bound.

For example, $B = \{2, 4, 6, \ldots\}$ is bounded from below. Here lower bound is 2 since for all $x \in B$, $x \geq 2 = M$.

Note: Finite sets are always bounded

Exercise

State whether each set of real numbers is bounded or unbounded:

a) $A = \{x : x < 5\}$ ✓

It is unbounded. It is bounded from above

b) $B = \{\ldots, -10, -5, 0, 5, 10, \ldots\}$

Unbounded. No lower or upper bound is present

c) $C = \{2^{18}, 3^{-7}, 7, 0, 8^{35}\}$

Bounded.

d) $D = \{2, 4, 8, \ldots, 2^n, \ldots\}$

It is unbounded. It is bounded from below.

Exercise

e) $E = \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots, (\frac{1}{2})^n, \ldots\}$

It is bounded because $E \subseteq I = [0,1]$.

f) $F = \{1, -1, \frac{1}{2}, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{3}, \ldots\}$ ✓ ✓

It is bounded because $F \subseteq [-1, 1]$.

Note Let $A = \{\dots, -10, -5, 0, 5, 10\}$

upper bound $(A) = 10$

Any number greater than 10 is also an upper bound of A. upper bound of $A = \{x : x \geq 10\}$

Least upper bound or supremum of $A = 10$

Let $B = \{2, 4, 6, \dots\}$

lower bound is (2) → greatest lower bound

so any real number less than 2 is also a lower bound of B.

Infimum

Supremum

Let A be a set of real numbers. A number M is called the least upper bound or supremum of A, written $M = sup(A)$ if M is an upper bound of A but any number less than M is not an upper bound of A, that is, for any positive number ϵ, there exist $a \in A$ such that, $M - \epsilon < a$.

For example, consider the set $\{\dots, -5, -10, 0, 5\} = A$

Here $sup(A) = 5$

let $\epsilon = 0.1$

$5 - 0.1 = 4.9$ is not an upper bound of A.

Completion Property of *R*

If a set of real numbers is bounded from above, then $sup(A)$ exists.

The real numbers R are said to be complete since it satisfies the above property. For example, consider the set:

$B = \{\dots, -2, -1, 0, 1, 2, 3, 4\}$

Here $sup(B) = 4$

Q is not Complete

Let A be the following subset of the rational numbers Q:

$A = \{x \in Q : x > 0, x^2 < 3\}$

$= (0, \sqrt{3})$

This set is bounded. Upper bound is $\sqrt{3}$

We cannot let $sup(A) = \sqrt{3}$ since $\sqrt{3}$ is not a rational number

Greatest Common Divisor

Definition

Suppose a and b are integers, not both 0. An integer d is called a common divisor of a and b if d divides both a and b, that is, if $d|a$ and $d|b$. If d is largest then it is denoted by $gcd(a, b)$ and it is called the greatest common divisor of a and b.

Example 1

a) The common divisor of 12 and 18 are $\pm 1, \pm 2, \pm 3, \pm 6$
Thus $\gcd(12, 18) = 6$. Similarly,
$\gcd(12, -18) = 6$, $\gcd(12, -16) = 4$,
$\gcd(29, 15) = 1$, $\gcd(14, 49) = 7$

b) For any integer a, we have $\gcd(1, a) = 1$.

Example 2

a) For any prime p, we have $\gcd(p, a) = p$ if $p \mid a$. Similarly $\gcd(p, a) = 1$ if $p \nmid a$.
For example, $\gcd(5, 10) = 5$,
and $\gcd(5, 11) = 1$

Important Result

Suppose $d = gcd(a, b)$. Then there exist integers x and y such that $d = ax + by$.

For example, $\gcd(12, 18) = 6 = d$
So, $d = ax + by$
$6 = 12x + 18y$
$6 = 12(-1) + 18(1)$
Hence, $x = -1$, $y = 1$

Theorem

A positive integer $d = gcd(a, b)$ if and only if d has the following properties:

1) d divides both a and b.
2) If c divides both a and b, then $c \mid d$.

$\gcd(12, 18) = 6$
Here $d = 6$ divides both 12 & 18.
Also 3 divides both 12 & 18.
Hence 3 divides 6.

Properties of the Greatest Common Divisor

1) $gcd(a, b) = gcd(b, a)$
2) If $x > 0$, then $gcd(ax, bx) = x.\ gcd(a, b)$.
$\gcd(12, 18) = \gcd(3 \cdot 4, 3 \cdot 6)$
$= 3 \cdot \gcd(4, 6)$
$= 3 \cdot 2 = 6$
3) If $d = gcd(a, b)$, then $gcd(a/d, b/d) = 1$.
$\gcd(12, 18) = \gcd\left(\frac{12}{6}, \frac{18}{6}\right) = \gcd(2, 3) = 1$
4) For any integer x, $gcd(a, b) = \gcd(a, b + ax)$.
$\gcd(12, 18) = \gcd(12, 18 + 12(2))$
$= \gcd(12, 42) = 6$

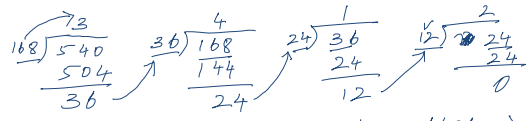
Euclidean Algorithm

In this topic, we will find very efficient algorithm for finding both $d = gcd(a, b)$ and the integers x and y such that $d = ax + by$.

This algorithm, called the Euclidean algorithm, consists of repeatedly applying the division algorithm(long division). This algorithm is illustrated with the following example.

Example 1

Let $a = 540$ and $b = 168$. We find $d = gcd(a, b)$ by dividing a by b and then repeatedly dividing each remainder into the divisor until obtaining a zero remainder.



$gcd(540, 168) = gcd(168, 36) = gcd(36, 12)$
$= gcd(12, 24) = 12.$

$12 = 540x + 168y$

Example 1

Next we find x and y such that $d = 540x + 168y$. The first three quotients from the previous long division gives the equations:

(1) $540 = 3(168) + 36 \Rightarrow 36 = 540 - 3(168)$

(2) $168 = 4(36) + 24 \Rightarrow 24 = 168 - 4(36)$

(3) $36 = 1(24) + 12 \Rightarrow 12 = 36 - 1(24)$

Here
$12 = 36 - 1(24) = 36 - 1((168 - 4(36))$
$= 36 - 1(168) + 4(36) = 5(36) - 1(168)$
$= 5(540 - 3(168)) - 1(168)$
$= 5(540) - 15(168) - 1(168)$
$12 = 5(540) - 16(168)$
$= 5a - 16(b)$

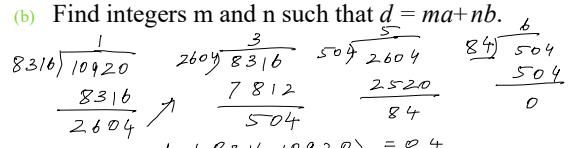
Hence $x = 5$, $y = -16$

Example 2

Let $a = 8316$ and $b = 10920$.

(a) Find $d = gcd(a, b)$.

(b) Find integers m and n such that $d = ma + nb$.



Here $gcd(8316, 10920) = 84$

Here (1) $10920 = 1(8316) + 2604 \Rightarrow 2604 = 10920 - 1(8316)$

(2) $8316 = 3(2604) + 504 \Rightarrow 504 = 8316 - 3(2604)$

(3) $2604 = 1(504) + 84 \Rightarrow 84 = 2604 - 1(504)$

Example 2

Let $a = 8316$ and $b = 10920$.

(b) Find integers m and n such that $d = ma + nb$.

$84 = 2604 - 5(504)$
$= 2604 - 5[8316 - 3(2604)]$
$= 2604 - 5(8316) + 15(2604)$
$= 16(2604) - 5(8316)$
$= 16(10920 - 1(8316)) - 5(8316)$
$= -21(8316) + 16(10920)$

$d = -21a + 16b$

Hence $m = -21$ and $n = 16$

Least Common Divisor Multiple

Suppose a and b are nonzero integers. Note that $|ab|$ is a positive common multiple of a and b. Thus there exists a smallest positive common multiple of a and b; it is denoted by $lcm(a, b)$ and it is called the least common multiple of a and b.

Example 3

a) $lcm(2, 3) = 6, lcm(4, 6) = 12, lcm(9, 10) = 90.$

$2 = 2, 4, (6), 8, 10, 12$

$3 = 3, (6), 12, 15$

b) For any positive integer a, we have $lcm(1, a) = a$.

$(5), 10, 15$

$1, 2, 3, 4, (5), 6 \ldots \quad lcm(1, 5) = 5$

c) For any prime p and any positive integer a, $lcm(p, a) = a$ or $lcm(p, a) = ap$ according as $p \mid a$ or $p \nmid a$.

Let $lcm(5, 10) = 5$

$lcm(5, 11) = 5 \cdot 11 = 55$

Theorem

Suppose a and b are nonzero integers. Then

$$lcm(a,b) = \frac{|ab|}{\gcd(a,b)}$$

Let $a = 8316$ and $b = 10920$. Find lcm(a, b).

As we know $gcd(8316, 10920) = 84$. So

$$lcm(a, b) = \frac{(8316)(10920)}{84}$$

$$= 1081080$$

Fundamental Theorem of Arithmetic

Theorem

Every integer $n>1$ can be expressed uniquely as a product of primes.

$$n = p_1^{m_1} p_2^{m_2} \cdots p_r^{m_r}$$

where m_i are positive and $p_1 < p_2 < \cdots < p_r$.

Example 1

Find the unique factorization of each number:

a) 135 (b) 1330 (c) 211

Sol: (a)

$135 = 5 \times 3 \times 3 \times 3 = 3^3 \times 5$

5	135
3	27
3	9
3	3
	1

b) $1330 = 2^1 \times 5^1 \times 7^1 \times 19^1$

2	1330
5	665
7	133
19	19
	1

c) 211 is a prime, so it cannot be factored.

Example 1

Let $a = 2^3 \cdot 3^5 \cdot 5^4 \cdot 11^6 \cdot 17^3$ and $b = 2^5 \cdot 3^3 \cdot 7^2 \cdot 11^4 \cdot 13^2$

Find gcd(a, b) and lcm(a, b).

Sol: $gcd(a, b) = 2^3 \cdot 3^3 \cdot 11^4 = ?$

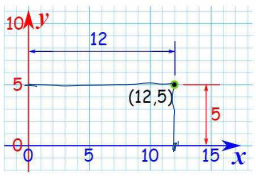
$lcm(a, b) = 2^5 \cdot 3^5 \cdot 5^4 \cdot 7^2 \cdot 11^6 \cdot 13^2 \cdot 17^3$

$= ?$

Product Sets

Ordered Pairs

Two numbers written in a certain order. Usually written in parentheses like (12, 5) which can be used to show position on a graph, where the "*x*"(horizontal) value is first and the "*y*"(vertical) value is second. So (12, 5) is 12 units along, and 5 units up.



$(a,b) \neq (b,a)$

$(a,b) = (c,d)$ iff $a=c, b=d$

Product Sets

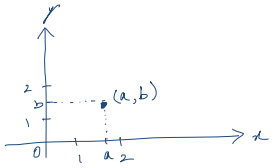
Let A and B be two sets. The product set or cartesian product of A and B, written $A \times B$ is the set of all ordered pairs (a, b) such that $a \in A$ and $b \in B$. Namely

$$A \times B = \{(a,b) : a \in A, b \in B\}$$

One usually writes A^2 instead of $A \times A$.

Example 1

$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ is the set of all ordered pairs of real numbers. In *xy*-plane, each point P represents an ordered pair (*a, b*) of real numbers and vice versa. $\mathbb{R}^2$ is also called the Cartesian plane.



Example 2

Let $A = \{1, 2\}$ and $B = \{a, b, c\}$. Then

$A \times B = \{(1,a), (1,b), (1,c), (2,a), (2,b), (2,c)\}$

$B \times A = \{(a,1), (a,2), (b,1), (b,2), (c,1), (c,2)\}$

$A^2 = A \times A = \{(1,1), (1,2), (2,1), (2,2)\}$

$A \times B \neq B \times A$

$n(A \times B) = 6 = n(A) \cdot n(B)$

Product of Three or More Sets

The idea of a product of sets can be extended to any finite number of sets. Specifically, for any sets $A_1, A_2, \ldots, A_m$ the set of all *m*-element lists $(a_1, a_2, \ldots, a_m)$ where each $a_i \in A_i$ is called the cartesian product of the sets $A_1, A_2, \ldots, A_m$. It is denoted by $A_1 \times A_2 \times \ldots \times A_m$

Example 3

Let $A = \{1, 2\}$, $B = \{x, y, z\}$, $C = \{3, 4\}$. Find $A \times B \times C$.

$A \times B \times C = \{(a, b, c) : a \in A, b \in B, c \in C\}$

$A \times B \times C = \{(1,x,3), (1,x,4), (1,y,3), (1,y,4), (1,z,3), (1,z,4), (2,x,3), (2,x,4), (2,y,3), (2,y,4), (2,z,3), (2,z,4)\}$

$\therefore n(A \times B \times C) = 12 = n(A) \cdot n(B) \cdot n(C)$

Example 4

Let $A = \{1, 2\}$, $B = \{a, b, c\}$, $C = \{c, d\}$. Find $A \times (B \cap C)$.

Sol: $B \cap C = \{c\}$

$A \times (B \cap C) = \{1,2\} \times \{c\}$

$= \{(1,c), (2,c)\}$

Exercise: $A \times B = ?$

$B \times A = ?$

Relations

Definitions

Let A and B be sets. A binary relation or, simply, a relation from A to B is a subset of $A \times B$. Suppose R is a relation from A to B. Then R is a set of ordered pairs where each first element comes from A and each second element comes from B. That is, for each $a \in A$ and $b \in B$,

(i) $(a, b) \in R$, we say "a is related to b" written $a R b$

(ii) $(a, b) \notin R$, we say "a is not related to b, $a \not R b$.

Domain and Range

The domain of a relation R from A to B is the set of all first elements of the ordered pairs which belong to R, and so it is subset of A; and the range of R is the set of all second elements, and so it is a subset of B.

Note: Sometimes R is a relation from a set A to itself, that is, R is a subset of $A^2 = A \times A$. In such a case, we say that R is a relation on A.

Example 1

a) Let $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$, and let $R = \{(1,y), (1,z), (3,y)\}$. This is a relation from A to B.

$1 R y$, $1 R z$, $3 R y$ but

$1 \not R x$, $2 \not R x$, $2 \not R y$, $2 \not R z$, $3 \not R x$, $3 \not R z$

b) Suppose we say that two countries are adjacent if they have some part of their boundaries in common. Then "is adjacent to" is a relation R on the countries of the earth. Thus:

(India, Pakistan) $\in R$ but (Pak, Bangla) $\notin R$

(Iran, Pak) $\in R$

Example 1

c) Set inclusion $\subseteq$ is a relation on any collection of sets.
For, given any pair of sets A and B, either $A \subseteq B$ or $A \nsubseteq B$.

d) A familiar relation on the set Z of integers is " m divides n". We write $m|n$ when m divides n.
$(m,n) \in R$. Here $(6,30) \in R$ because $6|30$.
But $(7,25) \notin R$ because $7 \nmid 25$.

Universal, Empty and Equality Relation

Let A be any set. Then $A \times A$ and $\emptyset$ are subsets of $A \times A$ and hence are relations on A called the universal relation and empty relation, respectively. Thus

$$\emptyset \subseteq R \subseteq A \times A$$

An important relation on the set A is that of equality, that is, the relation

$$\{(a,a) : a \in A\}$$

$A = \{1, 2, 3\}$
$R = \{(1,1), (2,2), (3,3)\}$

which is usually denoted by $=$.

Inverse Relation

Let R be any relation from a set A to a set B. The inverse of R, denoted by R^{-1} is the relation from B to A. That is

$$R^{-1} = \{(b,a) : (a,b) \in R\}$$

For example if R = $\{(1, y), (1, z), (3, y)\}$ is the relation from $A = \{1, 2, 3\}$ to $B = \{x, y, z\}$, then

$$R^{-1} = \{(y,1), (z,1), (y,3)\}$$

Note: $(R^{-1})^{-1} = R$. Also domain of R^{-1} is the range of R, and vice versa.

Example 2

Let A = {1 , 2, 3, 4, 5, 6} , and let R be the relation on A defined by "*x divides y*", written $x|y$.

a) Write R as a set of ordered pairs.

b) Find the inverse relation R^{-1} of R.

Sol (a) $1|1, 1|2, 1|3, 1|4, 1|5, 1|6,$
$2|2, 2|4, 2|6, 3|3, 3|6, 4|4, 5|5, 6|6$

Hence, $R = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,2), (2,4), (2,6), (3,3), (3,6), (4,4), (5,5), (6,6)\}$

(b) $R^{-1} = \{(1,1), (2,1), (3,1), \cdots, (5,5), (6,6)\}$

Representation of Relations on Finite Sets

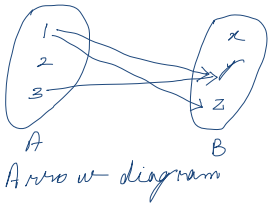
Consider, for example, the following relation R from $A = \{1, 2, 3\}$ to $B = \{x, y, z\}$:

$$R = \{(1, y), (1, z), (3, y)\}$$

We can represent this relation R in two ways as follows:

A \ B	x	y	z
1	0	1	1
2	0	0	0
3	0	1	0

matrix of the relation



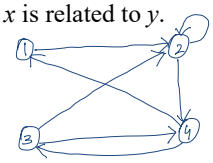
Arrow diagram

Directed Graphs of Relations on Sets

There is another way of picturing a relation R when R is a relation from a finite set A to itself. Consider the set A $= \{1, 2, 3, 4\}$ and the following relation R:

$R = \{(1, 2), (2,2), (2, 4), (3, 2), (3, 4), (4, 1), (4, 3)\}$

First we write down the elements of the set A, and then we draw an arrow from each element x to each element y whenever x is related to y.

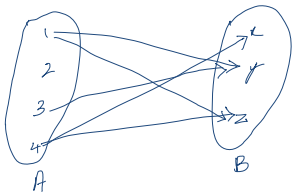


Example 1

Given $A = \{1, 2, 3 ,4\}$ and $B = \{x, y, z\}$. Let R be the following relation from A to B:

$R = \{(1, y), (1, z), (3, y), (4, x), (4, z)\}$

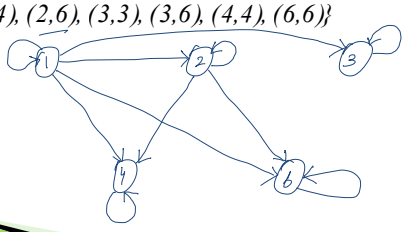
Draw the arrow diagram of R.



Example 2

Given $A = \{1, 2, 3 ,4, 6\}$ and let R be the relation on A defined by "x *divides* y", written $x|y$. Draw its directed graph.

Here $R = \{(1,1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2,2), (2,4), (2,6), (3,3), (3,6), (4,4), (6,6)\}$



Composition of Relations

Definition

Let A, B, C be sets, and let R be a relation from A to B and let S be a relation from B to C. Then R and S give rise to a relation from A to C denoted by $R \circ S$ and defined as follows:

$R \circ S = \{(a, c)$: *there exist* $b \in B$ *for which* $(a, b) \in R$ *and* $(b, c) \in S\}$

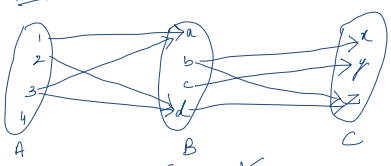
That is, $a(R \circ S)c$ whenever there exist $b \in B$ such that aRb and bSc.

This relation $R \circ S$ is called the composition of R and S; it is sometimes denoted by RS.

Example 1

Let $A = \{1,2,3,4\}$, $B = \{a,b,c,d\}$, $C = \{x,y,z\}$, and let $R = \{(1,a), (2,d), (3,a), (3,d)\}$ and $S = \{(b,x), (b,z), (c,y), (d,z)\}$. Then find $R \circ S$.

$S \circ R = ?$



$R \circ S = \{(2,z), (3,z)\}$

Note

Suppose R is a relation on a set A. Then $R \circ R$, the composition of R with itself, is always defined, and $R \circ R$ is sometimes denoted by R^2. Similarly $R^3 = R^2 \circ R = R \circ R \circ R$. Thus R^n is defined for all positive n.

$R^4 = R^3 \circ R$

$R^n = R^{n-1} \circ R = R \circ R \circ \cdots \circ R$ (n times)

Composition of Relations

Definition

Let A, B, C be sets, and let R be a relation from A to B and let S be a relation from B to C. Then R and S give rise to a relation from A to C denoted by $R \circ S$ and defined as follows:

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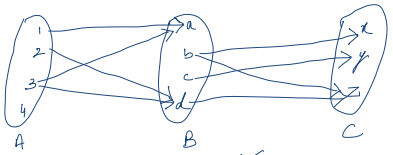
That is, $a(R \circ S)c$ whenever there exist $b \in B$ such that aRb and bSc.

This relation $R \circ S$ is called the composition of R and S; it is sometimes denoted by RS.

Example 1

Let $A = \{1,2,3,4\}$, $B = \{a,b,c,d\}$, $C = \{x,y,z\}$, and let $R = \{(1,a), (2,d), (3,a), (3,d)\}$ and $S = \{(b,x), (b,z), (c,y), (d,z)\}$. Then find $R \circ S$.

$S \circ R = ?$



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$R^4 = R^3 \circ R$

$R^n = R^{n-1} \circ R = R \circ R \circ \cdots \circ R$ (n times)

Types of Relations

Reflexive Relation

Consider a given set A. A relation R on a set A is reflexive if $a \; R \; a$ for every $a \in A$, that is, $(a,a) \in R$ for every $a \in R$.

For example, let $A = \{1, 2, 3, 4\}$ and consider the relation $R = \{(1,1), (1, 2), (2,1), (2, 2), (3, 3), (4, 4)\}$.

$S = \{(1,1), (1,2), (2,2), (3,3)\}$ non reflexive

Symmetric Relation

A relation R on a set A is symmetric if whenever $a \; R \; b$ then $b \; R \; a$, that is, if whenever $(a, b) \in R$, then $(b, a) \in R$.

For example, let $A = \{1, 2, 3, 4\}$ and consider the relation $R = \{(1,1), (1, 2), (2,1), (2, 2), (3, 3), (4, 4)\}$.

Antisymmetric Relation

A relation R on a set A is antisymmetric if whenever $a \; R \; b$ and $b \; R \; a$ then $a = b$, that is, if whenever $(a, b) \in R$, and $(b, a) \in R$, then $a = b$ or if $(a, b) \in R$ with $a \neq b$, then (b, a) should not belong to R.

For example, let $A = \{1, 2, 3, 4\}$ and consider the relation $R = \{(1,1), (1, 2), (2, 3), (1, 3), (4, 4)\}$.

Transitive Relation

A relation R on a set A is transitive if whenever $a \; R \; b$ and $b \; R \; c$ then $a \; R \; c$, that is, if whenever $(a, b) \in R$, and $(b, c) \in R$, then $(a, c) \in R$.

For example, let $A = \{1, 2, 3, 4\}$ and consider the relation $R = \{(1, 3), (2, 1)\}$ and $S = \{(1,1), (1, 2), (2, 3), (1, 3), (4, 3)\}$.

$S^2 = S \circ S \subseteq S$

Example 1

Consider the following relations:

a) Relation $\leq$ on the set Z of integers.
b) Set inclusion $\subseteq$ on a collection of sets.
c) Relation $|$ of divisibility on the set P of positive integers.

Reflexive $a \leq a$ for all $a \in Z$, $2 \leq 2$, $-2 \leq 2$ etc
Relation (a) is reflexive.
Relation (b) is reflexive
Relation (c) is reflexive.

$A \subseteq A$
$A R A$
$2|2$, $3|3$

Example 1

Symmetric $3 \leq 4$, but $4 \not\leq 3$, so relation (a) is not symmetric
If $A \subseteq B$, then $B \not\subseteq A$. $\{1,2\} \subseteq \{1,2,3\}$ but $\{1,2,3\} \not\subseteq \{1,2\}$. so relation (b) is not symmetric.
$2|6$ but $6 \nmid 2$. so relation (c) is not symmetric.

Antisymmetric
If $a \leq b$ and $b \leq a$, then $a = b$. so relation (a) is antisymmetric.
If $A \subseteq B$ and $B \subseteq A$, then $A = B$. So relation (b) is antisymmetric.
If $m|n$ and $n|m$, then $n = m$. so relation (c) is also antisymmetric.

Transitive (a) If $a \leq b$ and $b \leq c$, then $a \leq c$. $2|4, 4|8$ then $2|8$.
(b) If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$
(c) If $a|b$, and $b|c$, then $a|c$.

Remark

The properties of being symmetric and antisymmetric are not negative of each other. For example, the relation R = {(1, 3), (3, 1), (2, 3)} is neither symmetric and antisymmetric . On the other hand, the relation S = {(1, 1), (2, 2)} is both symmetric and antisymmetric.

(a, b)

① A relation R is transitive iff $R^n \subseteq R$ for every $n \geq 1$.

$R^2 = R \circ R$

$R^2 \subseteq R$, $R^3 \subseteq R$, ...

Example 2

Let A = {1, 2, 3, 4}. Consider the following relation R on A:

R = {(1, 1), (2, 2), (2, 3), (3, 2), (4, 2), (4, 4)}.

Reflexive: R is not reflexive because $3 \in A$ but $(3,3) \notin R$.

Symmetric: R is not symmetric because $(4,2) \in R$ but $(2,4) \notin R$.

Antisymmetric: R is not antisymmetric because $(2,3) \in R$ and $(3,2) \in R$ but $2 \neq 3$.

Transitive: Here $(4,2) \in R$ and $(2,3) \in R$ but $(4,3) \notin R$ so R is not transitive.

Closure of Relations

Definition

Consider a relation *R* on a set *A*. *R* may or may not have property *P*, such as reflexivity, symmetry, or transitivity.

If there is a relation *P(R)* with property *P* containing *R* such that *P(R)* is the subset of every relation with property *P* containing *R*, then *P(R)* is called the closure of *R* with respect to *P*.

$R \subseteq P(R)$

Notation

The reflexive, symmetric and transitive closures of a relation *R* will be denoted respectively by:

reflexive(*R*), symmetric(*R*), *transitive(R)*

The next theorem tells us how to easily obtain the reflexive and symmetric closures of a relation.

Theorem 1

Let *R* be a relation on a set *A*. Then:

(i) $R \cup \Delta_A$ is the reflexive closure of *R*.

(ii) $R \cup R^{-1}$ is the symmetric closure of *R*.

here $\Delta_A = \{(a,a) : a \in A\}$ is the equality relation on *A*.

Example 1

Consider the following relation R on a set $A = \{1, 2, 3, 4\}$:

$R = \{(1,1), (1,3), (2,4), (3,1), (3,3), (4,3)\}$

Then $reflexive(R) = R \cup \Delta_A$

$= R \cup \{(2,2), (4,4)\}$

$= \{(1,1), (1,3), (2,4), (3,1), (3,3), (4,3), (2,2), (4,4)\}$ ✓

$Symmetric(R) = R \cup R^{-1}$

$= R \cup \{(4,2), (3,4)\}$

$= \{(1,1), (1,3), (2,4), (3,1), (3,3), (4,3), (4,2), (3,4)\}$

Theorem 2

Let R be a relation on a set A with n elements. Then

$$transitive(R) = R \cup R^2 \cup \ldots \cup R^n$$

Here $R^2 = R \circ R$, $R^3 = R^2 \circ R$, ..., $R^n = R^{n-1} \circ R$

Example 2: Consider the following relation on A = {1, 2, 3}:

$$R = \{(1,2), (2,3), (3,3)\}$$

Then $transitive(R) = R \cup R^2 \cup R^3$

Here $R^2 = R \circ R = \{(1,2), (2,3), (3,3)\} \circ \{(1,2), (2,3), (3,3)\}$

$= \{(1,3), (2,3), (3,3)\}$

$R^3 = R^2 \circ R = \{(1,3), (2,3), (3,3)\}$

So, $transitive(R) = R \cup R^2 \cup R^3$

$= \{(1,2), (2,3), (3,3), (1,3)\}$

Partitions

Definition

Let S be a nonempty set. A partition of S is a subdivision of S into nonoverlapping, nonempty subsets. Precisely, a partition of S is a collection $P = \{A_i\}$ of nonempty subsets of S such that

(i) Each $a \in S$ belongs to one of the A_i

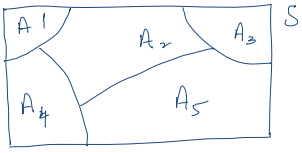
(ii) The sets $\{A_i\}$ are mutually disjoint; that is,

If $A_i \neq A_j$, then $A_i \cap A_j = \emptyset$

The subsets in a partition are called cells. Thus each $a \in S$ belongs to exactly one of the cells.

Venn Diagram of Partition

Following figure is a Venn diagram of a partition of the rectangular set S of points into five cells:



System of Representative

Given a partition $P = \{A_i\}$ of a set S, any element $b \in A_i$ is called a representative of the cell A_i and a subset B of S is called a system of representatives if B contains exactly one element of each of the cells of P.

Example 1

Consider the following collections of subsets of S = {1, 2,…, 9}:

(i) $P_1 = [\{1,3,5\}, \{2,6\}, \{4,8,9\}]$ Not a partition

(ii) $P_2 = [\{1,3,5\}, \{2,4,6,8\}, \{5,7,9\}]$ Not a partition

(iii) $P_3 = [\{1,3,5\}, \{2,4,6,8\}, \{7,9\}]$ This is a partition

Note that B = {1, 2, 7} is a system of representative of the partition given in (iii).

B = {3, 2, 9} is also system of representative

Example 2

Find all partition of the set S = { a, b, c} .

Sol: Note that we can make partitions of S containing 1, 2 or 3 cells.

(1) [S]

(2) $[\{a\}, \{b,c\}], [\{b\}, \{a,c\}], [\{c\}, \{a,b\}]$

(#3) $[\{a\}, \{b\}, \{c\}]$

Equivalence Relations

Definition

Let S be a nonempty set. A relation R on S is an equivalence relation if R is reflexive, symmetric and transitive. That is, R is an equivalence relation on S if it has the following three properties:

1) For every $a \in S$, aRa
2) If aRb, then bRa.
3) If aRb and bRc, then aRc.

The general idea behind an equivalence relation is that it is a classification of objects which are in some way alike. We use ~ or ≡ to denote the equivalence relation.

Example 1

The relation = of equality on any set S is an equivalence relation. How?

(1) $a = a$ for every $a \in S$.

(2) If $a = b$, then $b = a$.

(3) If $a = b$ and $b = c$, then $a = c$.

Example 2

The relation $\subseteq$ of set inclusion is not an equivalence relation. It is reflexive and transitive, but it is not symmetric since

$A \subseteq A$, ✓

If $A \subseteq B$, $B \subseteq C$, $A \subseteq C$. ✓

If $A \subseteq B$, then $B \not\subseteq A$.

Example 3

Let m be a fixed positive integer. Two integers a and b are said to be *congruent modulo m* written $a \equiv b \pmod{m}$ If m divides $a - b$. For example, if $m = 4$, then $11 \equiv 3 \pmod 4$ since $4 \mid 11 - 3$, $22 \equiv 6 \pmod 4$

We claim that this relation of congruence modulo m is an equivalence relation.

(1) $a \equiv a \pmod m$ for every $a \in Z$, because $m \mid a - a = 0$.

(2) If $a \equiv b \pmod m$, then $m \mid a - b \Rightarrow m \mid -(a-b)$
$\Rightarrow m \mid b - a$
$\Rightarrow b \equiv a \pmod m$

Example 3

(3) If $a \equiv b \pmod m$ and $b \equiv c \pmod m$, then we will show that $a \equiv c \pmod m$

Here m divides $a - b$ and m divides $b - c$.

So, m divides $(a - b) + (b - c)$.

// // $a - c$.

Hence, $a \equiv c \pmod m$.

Equivalence Relations and Partition

Equivalence Class and Quotient Set

Suppose R is an equivalence relation on a set S. For each a in S, let $[a]$ denote the set of elements of S to which a is related under R; that is,

$$[a] = \{x: (a, x) \in R\}$$

We call $[a]$ the equivalence class of a in S under R. The collection of all such equivalence classes is denoted by S/R, that is,

$$S/R = \{[a] : a \in S\}$$

It is called the quotient set of S by R.

Theorem 1

Let R be an equivalence relation on a set S. Then the quotient set S/R is a partition of S. Conversely, suppose $P = \{A_i\}$ is a partition of a set S. Then there is an equivalence relation $\sim$ on S such that the set $S/\sim$ of equivalence classes is the same as the partition $P = \{A_i\}$.

Example 1

Consider the relation $R = \{(1,1), (2,2), (1,3), (3,1), (3,3), (4,4)\}$ on $S = \{1, 2, 3, 4\}$.

Observe that R is an equivalence relation.

$[1] = \{1,3\}$, $[2] = \{2\}$, $[3] = \{1,3\}$, $[4] = \{4\}$

Note that $[1] = [3]$. Now,

$S/R = \{[1], [2], [4]\}$ is a partition of S.

System of representative $\{1,2,4\}$ or $\{2,3,4\}$

Example 2

$3 \equiv 0 \pmod 3$

$3 \equiv 6 \pmod 3$

Let R_3 be the relation on the set Z of integers defined by

$$x \equiv y \pmod 3$$

Here $x \equiv y \pmod 3$ is an equivalence relation

It means $x - y$ is divisible by 3.

$[0] = \{\ldots, -6, -3, 0, 3, 6, 9, \ldots\}$

$[1] = \{\ldots, -5, -2, 1, 4, 7, 10, \ldots\}$

$[2] = \{\ldots, -7, -4, -1, 2, 5, 8, 11, \ldots\}$

$[3] = \{\ldots, -6, -3, 0, 3, 6, 9, \ldots\}$

$Z/R_3 = \{[0], [1], [2]\}$ is a partition of Z.

$2 \equiv -1 \pmod 3$

$2 \equiv -4 \pmod 3$

$0 \equiv 3 \pmod 3$

$0 \equiv 6 \pmod 3$

$0 \equiv -3 \pmod 3$

$1 \equiv 4 \pmod 3$

$1 \equiv 7 \pmod 3$

$1 \equiv -2 \pmod 3$

$1 \equiv -5 \pmod 3$

$2 \equiv 5 \pmod 3$

$2 \equiv 8 \pmod 3$

Example 2

$Z = [0] \cup [1] \cup [2]$

$Z/R_5 = Z/5Z = Z_5$

$\{0, 1, 2\}$ is a system of representatives of the equivalence classes.

Proof of Theorem 1

For the first part, we want to prove that

(i) For each $a \in S$, we have $a \in [a]$.

(ii) $[a] = [b]$ if and only if $(a, b) \in R$.

(iii) If $[a] \neq [b]$, then $[a]$ and $[b]$ are disjoint.

Proof of Theorem 1

Proof of Theorem 1

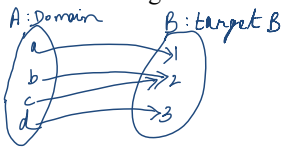
For the 2nd part, let $a, b \in S$ and define $a \sim b$ if a and b belong to the same cell A_k in P. We need to show that $\sim$ is reflexive, symmetric and transitive.

Proof of Theorem 1

Functions

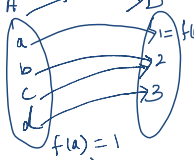
Functions

Suppose that to each element of a set A we assign a unique element of a set B; the collection of such assignments is called a function from A into B. The set A is called the domain of the function, and the set B is called the target set.



Notation

Suppose $f: A \to B$ and $a \in A$. Then $f(a)$ will denote the unique element of B which f assigns to a.



f sends or maps a into $f(a) = 1$.

$Im(f)$ = image of f

$= \{1, 2, 3\}$ = range of f.

Here $f(b) = 2$, $f(c) = 2$, $f(d) = 3$

$Im(f) = \{b \in B$: there exists $a \in A$ for which $f(a) = b\}$

Note: $Im(f)$ is a subset of target set B.

Functions

Mathematical Formula:

$f(x) = x^2$, or $x \mapsto x^2$ (goes to)

$f(1) = 1^2 = 1$

$f(2) = 2^2 = 4$

$f(3) = 3^2 = 9$

Let $f(x) = y$, then $y = x^2$ → independent variable

dependent variable

Domain $f: A \to B$ → target set

domain

Functions

Remark: Suppose $f: A \to B$. If A' is a subset of A, then $f(A')$ = set of images of elements in A'.

And if B' is a subset of B, then $f^{-1}(B')$ = set of elements of A each whose image belongs to B'.

$A' \subseteq A$, means $A' = \{a, b\}$

We call $f(A')$ the image of A', and $f^{-1}(B')$ the inverse image or preimage of B'.

$f(A') = \{1, 2\}$

Let $B' = \{1, 3\}$

so $f^{-1}(B') = \{a, c\}$

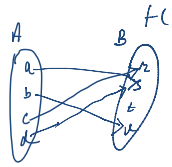
Examples

a) Let $f(x) = x^3$

The image of 2 is 8, $f(2) = 8$. Similarly $f(-3) = -27$.

b) Let $A = \{a, b, c, d\}$, $B = \{r, s, t, u\}$. Define a function from A to B as:

$f(a) = s$, $f(b) = u$, $f(c) = r$, $f(d) = s$



The image of $f = \{r, s, u\}$ = range

Examples

Identity Function

Consider any set A. Then there is a function from A into A which sends each element into itself. It is called the identity function on A and it is usually denoted by

$1_A : A \to A$ defined by

$1_A(a) = a$ for every element $a \in A$.

Functions as Relations

Functions as Relations

Every function $f: A \to B$ give rise to a relation from A to B called the graph of f and defined by

Graph of $f = \{(a, b); a \in A, b = f(a)\}$

This ordered pair (a, b) is unique.

For example, consider the function $f(x) = x^2$. We can represent this function as a relation and draw its graph as follows:

$f(1) = 1$, $f(2) = 4$, $f(3) = 9$, $f(4) = 16$

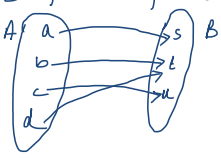
graph of $f = \{(1,1), (2,4), (3,9), (4,16)\}$

$= \{(x, y) : y = x^2\}$

Functions as Relations

set $A = \{a, b, c, d\}$, $B = \{s, t, u\}$

Define a function from A to B as:



graph of $f = \{(a, s), (b, t), (c, u), (d, t)\}$

$(b \not\to s)$

Here $f(a) = s$, $f(b) = t$,

$f(c) = u$, $f(d) = t$

Example 1

$A = \{a, b, c, d\}$, $B = \{s, u, r, s\}$

Let $f: A \to B$ be the function defined as follows:

$f(a) = s$, $f(b) = u$, $f(c) = r$, $f(d) = s$

graph of $f = \{(a,s), (b,u), (c,r), (d,s)\}$

Then the graph of f is the following set of ordered pairs:

Example 2

Consider the following relations on the set $A = \{1, 2, 3\}$.

$f = \{(1,3), (2,3), (3,1)\}$, $g = \{(1,2), (3,1)\}$,

$h = \{(1,3), (2,1), (1,2), (3,1)\}$

Here f is a function. Here $f(1) = 3$, $f(2) = 3$, $f(3) = 1$

g is not a function, since $2 \in A$ is not the first coordinate of any pair in g.

h is not a function, since $1 \in A$ appears as the first coordinate of two distinct ordered pairs.

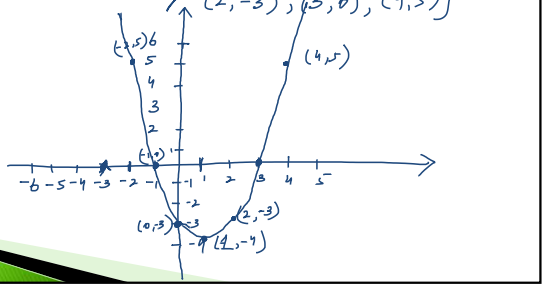
Example 3

Sketch the graph of $f(x) = x^2 + x - 6$.

x	$f(x)$
-2	5
-1	0
0	-3
1	-4
2	-3
3	0
4	5

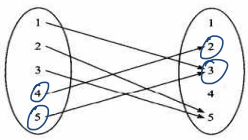
$f(-2) = (-2)^2 - 2 - 6 = 4 - 2 - 6 = -5$

graph of f: $\{(-2,5), (-1,0), (0,-3), (1,-4), (2,-3), (3,0), (4,5)\}$



Example 4

Let $A = \{1,2,3,4\}$ and let $f: A \to A$ be defined by the following diagram.



a) Find the graph of f. $\{(1,3), (2,5), (3,5), (4,2), (5,2)\}$

b) Find $f(A)$, the image of f. $f(A) = \{3, 5, 2\} = \{2, 3, 5\}$

c) Find $f(S)$ where $S = \{1, 3, 5\}$. $f(S) = \{3, 5\}$

d) Find $f^{-1}(T)$ where $T = \{2, 3\}$. $f^{-1}(T) = \{1, 4, 5\}$

Composition of Functions

Definition

Considers functions $f: A \to B$ and $g: B \to C$, that is, where the target set B of f is the domain of g. This relationship can be pictured by the following diagram:

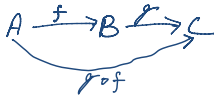
$a \in A \xrightarrow{f} B \xrightarrow{g} C$

Let $a \in A$, then $f(a) \in B$, $g(f(a)) \in C$

This new function is called the composition of f and g, and it is denoted by $g \circ f$.

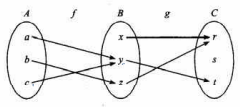
More briefly, if $f: A \to B$ and $g: B \to C$, then we define a new function $g \circ f: A \to C$ by

$(g \circ f)(a) = g(f(a))$



Example 1

Let $f: A \to B$ and $g: B \to C$ be the functions defined in the diagram.



We compute $g \circ f: A \to C$ by its definition:

$(g \circ f)(a) = g(f(a)) = g(r) = t$,

$(g \circ f)(b) = g(f(b)) = g(z) = r$,

$(g \circ f)(c) = g(f(c)) = g(r) = t$

Example 2

Let $f: \mathbf{R} \to \mathbf{R}$ and $g: \mathbf{R} \to \mathbf{R}$ be defined by $f(x) = x^2$, $g(x) = x+3$.

$(g \circ f)(2) = g(f(2)) = g(4) = 4+3 = 7$

$(f \circ g)(2) = f(g(2)) = f(5) = 5^2 = 25$

We compute general formula for these functions.

$(g \circ f)(x) = g(f(x)) = g(x^2) = x^2+3$

$(f \circ g)(x) = f(g(x)) = f(x+3) = (x+3)^2 = x^2+6x+9$

$f(2) = 4$

$g(2) = 2+3 = 5$

Example 3

Consider any function $f: A \to B$. Then one can easily show that $f \circ 1_A = f$ and $1_B \circ f = f$

identity function

$1_A : A \to A$ such that $1_A(a) = a$

check:

Let $f(a) = b$. Then

$(f \circ 1_A)(a) = f(1_A(a)) = f(a)$

$(1_B \circ f)(a) = 1_B(f(a)) = 1_B(b) = b = f(a)$

Theorem

Let $f: A \to B$, $g: B \to C$ and $h: C \to D$. Then prove that $(f \circ g) \circ h = f \circ (g \circ h)$.

Proof. Let $a \in A$. Then:

$((f \circ g) \circ h)(a) = (f \circ g)(h(a)) = f(g(h(a)))$

$(f \circ (g \circ h))(a) = f((g \circ h)(a)) = f(g(h(a)))$

Thus $((f \circ g) \circ h)(a) = (f \circ (g \circ h))(a)$ for every $a \in A$.

One to one, Onto and Invertible function

One-to-One Function

A function $f: A \to B$ is said to be one-to-one(injective) if different elements in the domain A have distinct images. Another way of saying the same things follows:

f is one-to-one if $f(a) = f(b)$ implies $a = b$.

For example, let $f: \mathbf{R} \to \mathbf{R}$ be defined by $f(x) = 2x-3$ is one-to-one. But $f(x) = x^2$ is not one-to-one.

$f(2) = 4$

$f(-2) = 4$

Here $f(2) = f(-2)$ but $2 \neq -2$.

Onto Function

A function $f: A \to B$ is said to be onto(surjective) function if every element of B is the image of some element in A or, in other words, if the image of f is the entire target set B. In such a case we say that f is a function of A onto B or that maps A onto B. That is:

f maps A onto B if $\forall\, b \in B, \exists\, a \in A$ such that $f(a) = b$.

For example, $f: \boldsymbol{R} \to \boldsymbol{R}$ be defined by $f(x) = 2x-3$ is onto.

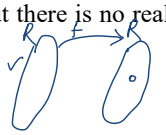
But $f(x) = 1 + x^2$ not onto since $0 \in \boldsymbol{R}$ but there is no real number x such that $f(x) = 0$.

$f(x) = 0$

$1 + x^2 = 0$

$x^2 = -1$

$x = \sqrt{-1} = i$



Invertible Function

A function $f: A \to B$ is said to be invertible if its inverse relation f^{-1} is a function from B to A. Equivalently, $f: A \to B$ is invertible if there exists a function $f^{-1}: B \to A$ called the inverse of f, such that

$f^{-1} \circ f = 1_A$ and $f \circ f^{-1} = 1_B$

Theorem

A function $f: A \to B$ is invertible if and only if f is both one-to-one and onto.

For example, let $f: \boldsymbol{R} \to \boldsymbol{R}$ be defined by $f(x) = 2x-3$. Now f is one-to-one and onto; hence f has an inverse function. We can find the inverse of f as follows:

$y = 2x - 3$

$\Rightarrow y + 3 = 2x$

$\Rightarrow \frac{y+3}{2} = x$

Now interchange x and y:

$y = \frac{x+3}{2}$

$\Rightarrow f^{-1}(x) = \frac{x+3}{2}$

$f \circ f^{-1}(x) = x$

$f^{-1} \circ f(x) = x$

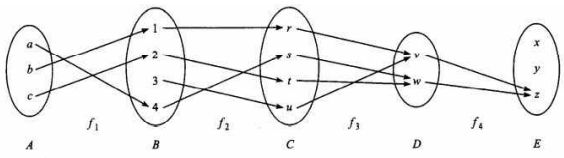
verify?

One-to-One Correspondence

If $f: A \to B$ is both one-to-one and onto, then f is called a one-to-one correspondence(bijective) between A and B. This terminology comes from the fact that each element of A will correspond to a unique element of B and vice versa. For example, $f(x) = 2x-3$ is a bijective function.

Example 1

Consider functions f_1, f_2, f_3 and f_4 defined in the following figure.



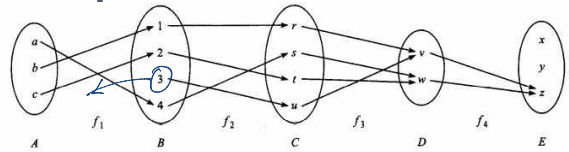
One-to-One

Here f_1 is one-to-one. f_2 is one-to-one.

Since $f_3(s) = f_3(t) = w$, so f_3 is not one-to-one.

Since $f_4(v) = f_4(w) = z_1$, so f_4 is not one-to-one.

Example 1



Onto Since $3 \in B$ is not the image under f_1 of any element of A, so f_1 is not onto.

image of $f_1 = \{1, 2, 4\} \neq$ target set B

Here f_2 is onto. Here f_3 is also onto.

Since image of $f_4 = \{z_1\} \neq$ target set E.

So f_4 is not onto. We can say that f_2 is invertible.

Example 2

Consider functions $f: A \rightarrow B$ and $g: B \rightarrow C$. Prove that if f and g are one-to-one, then the composition function $g \circ f$ is one-to-one.

Proof: Suppose $(g \circ f)(x) = (g \circ f)(y)$ | If $f(x) = f(y)$ then $x = y$.

$\Rightarrow g(f(x)) = g(f(y))$

$\Rightarrow f(x) = f(y)$

$\Rightarrow x = y$ ✓

Also you can prove that $g \circ f$ is onto.

Mathematical Functions

Integer Value Function

Let x be any real number. The integer value of x, written INT(x), converts x into an integer by deleting the fractional part of the number. Thus

INT(3.14) = 3, INT(7) = 7, INT($\sqrt{5}$) = 2,

INT(-8.5) = -8, INT(-3.33) = -3

Remainder Function: Modular Arithmetic

Let k be any integer and let M be a positive integer. Then

$k \pmod M = r$

(k *modulo* M) will denote the integer remainder when k is divided by M. More exactly, $k \pmod M$ is the unique integer r such that $k = Mq + r$ where $0 \le r < M$

When k is positive, simply divide k by M to obtain remainder r. Thus

$25 \pmod 7 = 4$, $25 \pmod 5 = 0$,

Here $k = 25$, $M = 7$ $35 \pmod{11} = 2$,

$3 \pmod 8 = 3$

$7\overline{)25}$ 3, 21, 4

Remainder Function: Modular Arithmetic

When k is negative, divide $|k|$ by the modulus M to obtain the remainder s. Then, when $s \neq 0$, $k \pmod M = M - s$. Thus:

$-26 \pmod 7 = 7 - 5 = 2$, $-371 \pmod 8 = 8 - 3 = 5$

$|-26| = 26$

$8\overline{)371}$ 46, 368, 3 = 5

$7\overline{)26}$ 3, 21, 5 = 5

$7\overline{)-26}$ -4, -28, 2

Remainder Function: Modular Arithmetic

The term "mod" is also used for the mathematical congruence relation, which is denoted and defined as follows:

$a \equiv b \pmod M$ if and only if M divides $b - a$

M is called the modulus, and $a \equiv b \pmod M$ is read "a is congruent to b modulo M"

$5 \equiv 5 \pmod 5$ because $\frac{5-5}{5} = \frac{0}{5} = 0$

$8 \equiv 16 \pmod 2$ because $\frac{8-16}{2} = \frac{-8}{2} = -4$

$5 \not\equiv 1 \pmod 3$

Note: ① $a \equiv a \pmod M$

② $a \pm M \equiv a \pmod M$

$8 \equiv 16 \pmod 2$

$8+2 \equiv 16 \pmod 2$

$8-2 \equiv 16 \pmod 2$

Arithmetic Operations with Modulo

Arithmetic modulo M refers to the arithmetic operations of addition, multiplication, and subtraction. It is quite simple: calculate as in ordinary arithmetic and reduce the result to the smallest positive remainder by dividing the modulus. For example,

$12+9 \equiv 21 \equiv 1 \bmod 5$ $\qquad$ $35 \times 7 \equiv 245 \equiv 0 \bmod 5$

$12-9 \equiv 3 \bmod 5$

$12+3 \equiv 0 \bmod 5$

$15-23 \equiv -8 \equiv 2 \bmod 5$ because $\frac{-8-2}{5} = -2$

Exponential and Logarithmic functions

Exponential Function

An exponential function is a mathematical in form as follows:

$$f(x) = a^x$$

where x is a variable and a is a constant which is called the base of the function and it should be greater than 0.

For example, $f(x) = 2^x$, $f(x) = 5^x$ etc are all exponential functions. $f(x) = 0.5^x$

$f(x) = -2^x$. Take $x = \frac{1}{2}$, then $f(x) = (-2)^{1/2} = \sqrt{-2}$ = undefined

Logarithmic Function

Logarithms are related to exponent as follows. Let b be a positive number. The logarithm of any positive number x to the base b, written

$$\log_b x$$

represents the exponent to which b must be raised to obtain x. That is,

$$y = \log_b x \text{ and } b^y = x$$

are equivalent statements.

Note: x cannot be negative and 0.

Examples

$\log_2 8 = 3$ since $2^3 = 8$, $\log_{10} 100 = 2$ since $10^2 = 100$

$\log_2 64 = 6$ since $2^6 = 64$, $\log_{10} 0.001 = -3$ since $10^{-3} = 0.001$.

For any base b,

1. $\log_b 1 = 0$ since $b^0 = 1$
2. $\log_b b = 1$ since $b^1 = 1$; $\log_2 2 = 1$

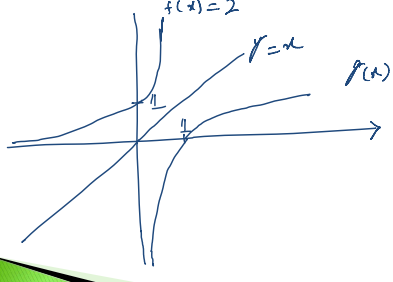
Common logarithms if it has base 10

Natural logarithms $\log_e x = \ln x$

Relationship Between the Exponential and Logarithmic Function

$f(x) = b^x$ and $g(x) = \log_b x$ is that are inverses of each other.

$(f \circ g)(x) = x$



Recursively Defined Functions

Definition

A recursive or inductive definition of a function consists of two steps.

1) Basis Step: Specify the value of the function at initial values. (e.g. $f(0)$ defined.)
2) Recursive Step: Give a rule for finding its values at an integer from its values at smaller integers. (For $n > 0$, define $f(n)$ in terms of $f(0), f(1), \ldots, f(n-1)$)

Example 1: Fibonacci Sequence

a) If $n = 0$ or $n = 1$, then $F_n = n$ $\quad F_0 = 0,\ F_1 = 1$
b) If $n > 1$, then $F_n = F_{n-2} + F_{n-1}$.

Here the base values are 0 and 1. Hence, the sequence can be found as:

$F_2 = F_0 + F_1 = 0 + 1 = 1$ $\quad 0, 1, 1, 2, 3, 5, 8, 13, \ldots$
$F_3 = F_1 + F_2 = 1 + 1 = 2$
$F_4 = F_2 + F_3 = 1 + 2 = 3$
$F_5 = F_3 + F_4 = 2 + 3 = 5$
$\vdots$

Example 2:

Find $f(1), f(2), f(3)$ and $f(4)$ if $f(n)$ is defined recursively by $f(0) = 1$ and $f(n+1) = f(n) + 2$ for $n = 0, 1, 2, \ldots$.

Sol: $f(n+1) = f(n) + 2$
put $n = 0$, $f(1) = f(0) + 2 = 1 + 2 = 3$
put $n = 1$, $f(2) = f(1) + 2 = 3 + 2 = 5$
put $n = 2$, $f(3) = f(2) + 2 = 5 + 2 = 7$
put $n = 3$, $f(4) = f(3) + 2 = 7 + 2 = 9$

$\{a, b, c, d\}$, $\{2, 3, 5, 7\}$, $\{x, y, z, t\}$

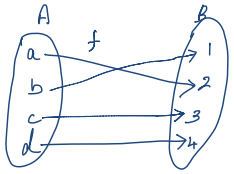
Equipotent Sets

Definition

Sets A and B are said to have the same cardinality or the same number of elements, or to be equipotent, written

$$A \approx B$$

if there is a function $f: A \to B$ which is bijective.



Here f is one-to-one
f is onto.
∴ f is bijective.
or f is one-to-one correspondence.

Theorem 1

The relation ≈ of being equipotent is an equivalence relation in any collection of sets. That is,

i) $A \approx A$ for any set A.

ii) If $A \approx B$, then $B \approx A$.

iii) If $A \approx B$ and $B \approx C$, then $A \approx C$.

Proof.

(i) Let $I_A : A \to A$ is a bijective mapping. It is defined as $I_A(a) = a$. $\forall\ a \in A$.

one-to-one: $I_A(a_1) = I_A(a_2) \Rightarrow a_1 = a_2$ so $A \approx A$.

Theorem 1

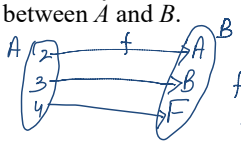
(ii) If $A \approx B$, there exists a bijective mapping $f : A \to B$. Hence $f^{-1} : B \to A$ is also a bijection. Therefore, $B \approx A$.

(iii) If $A \approx B$, $B \approx C$, then $f : A \to B$ and $g : B \to C$ are both bijective. Now, the composition $g \circ f : A \to C$ is also bijective. Hence $A \approx C$.

Example 1

Let A and B be sets with exactly three elements, say, $A = \{2, 3, 4\}$ and $B = \{\text{Anum, Bilal, Fahad}\}$.

Then clearly we can find the one-to-one correspondence between A and B.



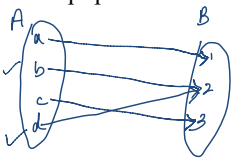
Hence $A \approx B$

$f(2) = \text{Anum}$, $f(3) = \text{Bilal}$, $f(4) = \text{Fahad}$

Remark: The same idea may be used to show that any two finite sets with the same number of elements are equipotent.

Example 2

Let $A = \{a, b, c, d\}$ and $B = \{1, 2, 3\}$. Then A and B are not equipotent.



This is not one-to-one.

Remark: In a similar way, we can say that any two finite sets with different number of elements are not equipotent.

Example 3

$I = [0, 1]$, $S = [a, b]$ are equipotent. $[1, 2]$ and $[3, 5]$

Let $I = [0, 1]$, the closed unit interval, and let S be any other closed interval, say $[a, b]$ where $a < b$. The function $f : I \to S$ defined by

$$f(x) = (b - a)x + a$$

$f(0) = a$, $f(1) = (b - a) + a = b$

is one-to-one and onto. Thus I and S are equipotent.

Remark: Any two closed intervals are equipotent by theorem 1.

$g(x) = (d - c)x + c$

Reason: Since $f : I \to S$ bijective so $f : S \to I$ is also bijective. Consider another closed interval T, say $[c, d]$. We can define bijective function $g : I \to T$ as in the above example. So by transitivity we can define another bijective function from S to T. Hence S and T are equipotent.

$f : S \to I$, $g : I \to T$, $g \circ f : S \to T$ is also bijective.

$f(x_1) = f(x_2)$

$(b - a)x_1 + a = (b - a)x_2 + a$

$\Rightarrow x_1 = x_2$

Example 3

Remark: Any two closed intervals are equipotent by theorem 1.

Reason: Since $f : I \to S$ bijective so $f : S \to I$ is also bijective. Consider another closed interval T, say $[c, d]$. We can define bijective function $g : I \to T$ as in the above example. So by transitivity we can define another bijective function from S to T. Hence S and T are equipotent.

Example 4

Consider the set $P = \{1, 2, 3, \ldots\}$ of positive integers and the set $E = \{2, 4, 6, \ldots\}$ of even positive integers. The following defines a one-to-one correspondence between P and E:

$P = \{1, 2, 3, \ldots\}$ $f(1)=2(1)=2$
$E = \{2, 4, 6, \ldots\}$ $f(2)=2(2)=4$

In other words, the function $f: P \to E$ defined by $f(n) = 2n$ is one-to-one and onto. Thus P and E have the same cardinality.

Note: What does this example show us? $E \subseteq P$

Definition

A set S is infinite if it has the same cardinality as a proper subset of itself. Otherwise S is finite.

For example, the counting numbers P, the natural numbers(nonnegative integers) N, the integers Z, the rational numbers Q, and the real numbers R.

$A = \{1, 2, 3, 4\}$
$B = \{1, 2, 3\}$
$B \subset A$

Denumerable and Countable Sets Part 1

Denumerable Set

A set D is said to be denumerable or countably infinite if D has the same cardinality as $P = \{1, 2, 3, \ldots\}$.

i.e $f: D \to P$ is bijective. E and P $f: P \to E$ defined by $f(n)=2n$ is bijective.

Countable Set

A set is countable if it is finite or denumerable, and a set is nondenumerable if it is not countable.

Note: Thus a set S is nondenumerable if S is infinite and S does not have the same cardinality as P.

Example 1

Any infinite sequence

$S = \{a_1, a_2, a_3, \ldots\}$

of distinct elements is countably infinite. or denumerable.

Let $f: P \to S$ defined by $f(n) = a_n$.

Here $f(1) = a_1$, $f(2) = a_2$, $f(3) = a_3$, ...

Similarly all the following sets are countably infinite.

$\{1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{n}, \ldots\}$, $\{1, -2, 3, -4, \ldots, (-1)^{n-1} n, \ldots\}$
$P = \{1\ 2\ 3\ 4 \ldots n \ldots\}$

$\{(1,1), (4,8), (9,27), \ldots, (n^2, n^3), \ldots\}$
$P = \{1\ 2\ 3 \ldots n\}$

Example 2

Consider the product set $P \times P$. The set $P \times P$ can be written as an infinite sequence as follows:

$(1,1)\ (1,2)\ (1,3)\ (1,4) \ldots$
$(2,1)\ (2,2)\ (2,3)\ (2,4) \ldots$
$(3,1)\ (3,2)\ (3,3)\ (3,4) \ldots$
$(4,1)\ (4,2)\ (4,3)\ (4,4)$
$\vdots\ \ \vdots\ \ \vdots\ \ \vdots$ $P \times P \leftrightarrow P$

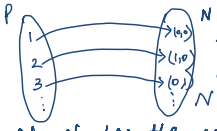
$P \times P = \{(1,1), (2,1), (1,2), (1,3), (2,2), (3,1), \ldots\}$
$P = \{1\ 2\ 3\ 4\ 5\ 6 \ldots\}$

Example 3

Recall that $N = \{0, 1, 2, ...\}$ is the set of natural numbers or nonnegative numbers. Now each positive integer $a \in P$ can be written uniquely in the form

$a = 2^r(2s+1)$, $r, s \in N$

consider the function $f: P \to N \times N$ defined by $f(a) = (r, s)$.



This f is one-to-one and onto. Thus $N \times N$ is denumerable.

$N \times N$ has the same cardinality as P.

Note: $P \times P$ is a subset of $N \times N$.

$r = s = 0$, $a = 2^0(2\cdot0+1) = 1(0+1)$, $a = 1$

put $r = 1$, $s = 0$, $a = 2^1(2\cdot0+1) = 2$

put $r = 0$, $s = 1$, $a = 2^0(2\cdot1+1) = 1(3) = 3$

Example 4

The set Z of integers can be put into a one-to-one correspondence with $P = \{1, 2, 3, ...\}$ as follows:

The formula for the function $f: P \to Z$ is given by:

Denumerable and Countable Sets Part 2

Theorem 1

Every infinite set contains a subset which is denumerable.

Proof. Let A be an infinite set. Suppose $f: P(A) \to A$. consider the following sequence.

$a_1 = f(A)$

$a_2 = f(A \setminus \{a_1\})$

$a_3 = f(A \setminus \{a_1, a_2\})$

$a_n = f(A \setminus \{a_1, a_2, ..., a_{n-1}\})$

Since A is infinite, $A \setminus \{a_1, ..., a_{n-1}\}$ is not empty for every $n \in P$. Furthermore $a_n \neq a_i$ for $i < n$.

Therefore, $D = \{a_1, a_2, ...\}$

Here, $D \subseteq A$

Also D is denumerable.

Theorem 2

P is denumerable and E is a subset of P.

Also $E \approx P$, means?

A subset of a denumerable set is finite or denumerable.

Proof: Let $A = \{a_1, a_2, ...\}$ is a denumerable set. Let $B \subseteq A$. If $B = \emptyset$, then B is finite. Suppose $B \neq \emptyset$. Let $b_1 \in B$ and b_1 be the first element in the sequence in A. Let $b_2 \in B$ and b_2 be the first element follows b_1 in the sequence A, so on. Then ~~$B = \{b_1, b_2, b_3, ...\}$~~ $B = \{b_1, b_2, b_3, ...\}$. If the sequence $b_1, b_2, ...$ ends, then B is finite. Otherwise B is denumerable.

Corollary

A subset of a countable set is countable.

Proof. Proof is obvious from theorem 2.

Theorem 3

Let $A_1, A_2, A_3, \ldots$ be a sequence of pairwise disjoint denumerable sets. Then the union

$$A_1 \cup A_2 \cup A_3 \cup \ldots = \cup(A_i : i \in P)$$

is denumerable.

Proof: Suppose $A_1 = \{a_{11}, a_{12}, a_{13}, \ldots\}$, $A_2 = \{a_{21}, a_{22}, a_{23}, \ldots\}$ $A_3 = \{a_{31}, a_{32}, a_{33}, \ldots\}$, ...

Then $S = A_1 \cup A_2 \cup A_3 \cup \cdots = \{a_{11}, a_{12}, a_{13}, \ldots, a_{21}, a_{22}, a_{23}, \ldots, a_{31}, a_{32}, a_{33} \ldots\}$

Define $D_n = \{a_{ij} : i+j = n, n > 1\}$. For example,

$i=1, j=1$, $n=2$, $D_2 = \{a_{11}\}$, $D_3 = \{a_{12}, a_{21}\}$, $D_4 = \{a_{13}, a_{22}, a_{31}\}$, ...

Note that each D_n is finite. Let $T = \cup\{D_j : j > 1\}$

Here T is countable. Also $T = A_1 \cup A_2 \cup A_3 \cdots$

Theorem 4

The set Q of rational numbers is denumerable.

Proof. Note that $Q = Q^+ \cup \{0\} \cup Q^-$. Let $f : Q^+ \to P \times P$ defined by $f(p/q) = (p, q)$ where $p/q \in Q^+$ expressed as the ratio of two relatively prime positive integers. Then f is one-to-one correspondence. Hence $Q^+ \approx P \times P$. As we know $P \times P \approx P$. so $Q^+ \approx P$. Similarly, Q^- is denumerable. The set Q of rational numbers, the union of Q^+, $\{0\}$, and Q^- is also denumerable.

Theorem 5

The set Z of integers is denumerable.

Proof: Let $f : P \to Z$ defined by

$$f(n) = \begin{cases} -\frac{n}{2} + \frac{1}{2} & \text{if } n \text{ is odd} \\ \frac{n}{2} & \text{if } n \text{ is even.} \end{cases}$$

For example,

$f(1) = -\frac{1}{2} + \frac{1}{2} = 0$

$f(2) = \frac{2}{2} = 1$

$f(3) = -\frac{3}{2} + \frac{1}{2} = -\frac{3+1}{2} = -1$

$f(4) = \frac{4}{2} = 2$

$\vdots$

$P = \{1 \quad 2 \quad 3 \quad 4 \quad 5 \cdots\}$

$\quad\;\; \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$

$Z = \{0 \quad 1 \quad -1 \quad 2 \quad -2 \cdots\}$

Here f is one-to-one. and f is onto. so f is bijective. Hence Z is denumerable.